SFPRM as an Operator-Split AKOrN: Exact Correspondences Between Spherical Fixed-Point Reasoning and Kuramoto Oscillatory Neurons

Anonymous Authors[1]

Abstract

Looped fixed-point reasoners and oscillator-based networks have developed separately, although both evolve unit vectors on spheres. We show that the Spherical Fixed-Point Reasoning Model (SFPRM) is, at the level of its update rule, a generalized network of Artificial Kuramoto Oscillatory Neurons (AKOrN) with zero natural frequency. Two elementary lemmas establish that normalized residual updates and spherical linear interpolation are exactly AKOrN Euler steps with state-dependent step sizes. Consequently, one SFPRM iteration is exactly an AKOrN step driven by the full recurrent map, its geodesic input injection and nGPT attention sub-layers are exactly AKOrN's stimulus and coupling steps, the iteration collapses to a single AKOrN step up to $O(\varepsilon^2)$ when step sizes are small, and the two forms share their linearization and hence SFPRM's damping window. We delineate where the models differ, read SFPRM's existence and convergence results as oscillator locking, and derive the locking condition of an oscillatory extension, showing also that a constant natural frequency can be absorbed into the recurrent block. The correspondence then suggests a model: an energy-based SFPRM whose recurrent map is a Riemannian gradient step on an explicit energy built from the geodesic injection, a symmetric attention energy with signed heads and hidden tokens, and a Hopfield memory. It inherits an energy that decreases at every iteration, convergence from any start with an a priori halting bound, a real Jacobian spectrum at halted states with the widest possible damping window, energy-based selection over random restarts, and training by conjugate gradient or equilibrium propagation. Reading the memory as a von Mises–Fisher mixture leads to a codebook variant whose readout is tied to answer prototypes and whose concentration is annealed over iterations. All identities are verified numerically. In a small Sudoku pilot the first energy model trains slowly and its energy does not rank solutions. The codebook variant matches SFPRM on 4×4 boards with one restart and lags it on 6×6, but its lowest-energy restart tracks the oracle on both, raising out-of-distribution accuracy from 74% to 99% on 4×4 and from 11% to 88% on 6×6 with 64 restarts.

1. Introduction

Looped, weight-tied models apply one block repeatedly and read their prediction from the state it converges to, so that depth becomes an iteration count chosen per input (Dehghani et al., 2019; Bai et al., 2019). Recent recurrent reasoners for puzzles such as Sudoku and mazes follow this pattern (Wang et al., 2025; Jolicoeur-Martineau, 2025), and the Fixed-Point Reasoning Model (FPRM) makes the loop an explicit damped fixed-point solve with residual-based halting (Movahedi et al., 2026). Its spherical counterpart, SFPRM (Anonymous, 2026), runs the same loop on the unit hypersphere of the normalized Transformer nGPT (Loshchilov et al., 2025): input injection and damping become geodesic operations, boundedness holds by construction, input injection guarantees a fixed point, and damping comes with an explicit step-size window.

A separate line of work builds networks from coupled oscillators. Artificial Kuramoto Oscillatory Neurons (AKOrN) generalize the Kuramoto model (Kuramoto, 1984) to unit vectors in N dimensions that rotate at natural frequencies, interact through learned couplings and are driven by a conditional stimulus (Miyato et al., 2025). Both families evolve unit vectors by moving along a tangent direction and renormalizing, yet the connection between them has not been made precise. A precise connection matters for two reasons: it lets results proved for one family (SFPRM's existence and damping guarantees, AKOrN's energy and synchronization picture) be read in the language of the other, and it identifies which oscillator mechanisms SFPRM already contains and

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

which it lacks.

Contributions.

- Two lemmas showing that normalized residual updates and spherical linear interpolation are exactly AKOrN Euler steps with state-dependent step sizes (Section 4).
- An exact equivalence: one SFPRM iteration is an AKOrN step with $\Omega = 0$, one oscillator per token and the full recurrent map as drive; geodesic injection and nGPT attention sub-layers are exactly AKOrN's stimulus and coupling steps, so an iteration is an operator-split AKOrN step (Section 5).
- A first-order collapse to a single AKOrN Euler step with explicit stimulus, coupling and self-drive, and an exact correspondence of linearizations that transfers SFPRM's damping window to the AKOrN form (Sections 5.3 and 5.4).
- A delineation of the differences between the models, an oscillator reading of SFPRM's theory, and an oscillatory extension with its locking condition, together with the observation that a constant natural frequency can be absorbed into the recurrent block (Sections 6 and 7).
- An energy-based SFPRM whose recurrent map is a gradient step on an explicit energy, with a Lyapunov and halting guarantee from any start, a real Jacobian spectrum at halted states, energy-based voting over restarts, and conjugate-gradient or equilibrium-propagation training, and a codebook variant whose energy ranks solutions; a pilot study measures what the energy costs and what it buys (Section 8).

Our analysis concerns update rules. All identities are verified numerically (Section 9), and the energy-based model is tested in a small controlled experiment. Table 1 summarizes the results.

Table 1. Summary of results. "Exact" statements hold for any step sizes satisfying stated positivity conditions.

Statement	Type	Where
Renormalized residual step = tangent step	exact	Lem. 4.1
Slerp = tangent step	exact	Lem. 4.2
SFPRM iteration = AKOrN step ($\Omega = 0$, $N = d$)	exact	Thm. 5.2
Injection = stimulus step	exact	Prop. 5.4
Attention sub-layer = coupling step	exact	Prop. 5.6
Iteration = composition of AKOrN steps	exact	Thm. 5.7
Collapse to a single AKOrN step	$O(\varepsilon^2)$	Thm. 5.8
Shared linearization and damping window	exact	Cor. 5.11
Constant Ω absorbed into the block	exact	Prop. 7.2
Locking condition with natural frequencies	exact	Prop. 7.3
Energy SFPRM: gradient = injection + symmetric coupling + memory	exact	Prop. 8.2
Energy descent, convergence, halting bound	exact	Thm. 8.3
Real spectrum, window $\eta < 2/\lambda_{\max}$	exact	Prop. 8.4
Energy selection over restarts	exact	Prop. 8.5

2. Related Work

Looped and fixed-point reasoners. Universal Transformers share weights across depth (Dehghani et al., 2019), and deep equilibrium models compute the fixed point of a weight-tied layer and differentiate through it implicitly (Bai et al., 2019). Hierarchical and tiny recursive models refine latent states with nested loops and deep supervision (Wang et al., 2025; Jolicoeur-Martineau, 2025). FPRM replaces the hierarchy by a single damped fixed-point loop with pre-norm, residual scaling and residual-based halting (Movahedi et al., 2026); SFPRM moves this loop to the hypersphere (Anonymous, 2026). Equilibrium Reasoners learn task-conditioned attractors and scale test-time compute through depth and random restarts (Huang et al., 2026), and Daulbaev et al. (2026) constrain the state to products of simplices and analyse the Jacobian at FPRM's halted states. We do not propose a new reasoner; we characterize SFPRM's update rule.

Normalized networks. nGPT keeps every hidden vector on the unit sphere and folds each sub-layer into the state by a normalized residual step with learned eigen learning rates (Loshchilov et al., 2025). SFPRM uses this block, and Lemma 4.1 is the observation that each such step is a tangent step.

Oscillatory networks. The Kuramoto model is the canonical model of synchronization (Kuramoto, 1984; Strogatz, 2000), and its non-Abelian generalizations evolve unit vectors on higher-dimensional spheres (Lohe, 2009). Oscillatory recurrent networks exploit oscillator dynamics for stable long-range memory (Rusch & Mishra, 2021) and for spatiotemporally structured representations through travelling waves (Keller & Welling, 2023). AKOrN builds feed-forward and recurrent networks from N-dimensional Kuramoto units (Miyato et al., 2025); Kuramoto phase encodings modulate attention in vision transformers (Xiao et al., 2026); and Continuous Thought Machines use the synchronization of neural activity as their representation (Darlow et al., 2025). Equilibrium propagation (Scellier & Bengio, 2017) has been extended to learn natural frequencies of Kuramoto networks at phase-locked equilibria

(Ahmadi, 2026).

3. Background

Section A gives a self-contained account of the mathematics used here: retractions and Riemannian gradients, rotations, gradient flows, damped fixed-point iterations, nGPT and operator splitting. We summarize the essentials.

3.1. Geometry of the Sphere

Let $\mathbb{S}^{n-1} = \{z \in \mathbb{R}^n : \|z\| = 1\}$. Every $v \in \mathbb{R}^n$ splits at z into a radial and a tangential part,

$$v = \langle v,\, z\rangle\, z + \mathrm{P}_z\, v, \qquad \mathrm{P}_z\, v := (\mathrm{I} - zz^\top)\, v, \quad (1)$$

where P_z projects onto the tangent space $\mathrm{T}_z\mathbb{S}^{n-1} = \{v : \langle v,\, z\rangle = 0\}$. We write $\Pi(y) := y/\|y\|$ for normalization and $\theta = \angle(u, v) := \arccos\langle u,\, v\rangle$. The exponential and logarithm maps are

$$\mathrm{Exp}_u(w) = \cos(\|w\|)\, u + \sin(\|w\|)\, \tfrac{w}{\|w\|}, \quad (2)$$

$$\mathrm{Log}_u(v) = \tfrac{\theta}{\sin\theta}\, \mathrm{P}_u\, v, \quad (3)$$

for $w \in \mathrm{T}_u\mathbb{S}^{n-1}$ and $\theta \in (0, \pi)$, and spherical linear interpolation is

$$\mathrm{Slerp}_\eta(u, v) = \frac{\sin((1-\eta)\theta)}{\sin\theta}\, u + \frac{\sin(\eta\theta)}{\sin\theta}\, v, \quad \eta \in [0, 1]. \quad (4)$$

For sequences $z = (z_1, \ldots, z_T)$ all operators act token-wise; $\odot$ is the element-wise product.

3.2. Kuramoto Oscillators and AKOrN

The Kuramoto model $\dot\theta_i = \omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i)$ becomes, for $x_i = (\cos\theta_i, \sin\theta_i)$ and the 90° rotation R,

$$\dot x_i = \omega_i R\, x_i + \mathrm{P}_{x_i}\Big(\sum_j K_{ij} x_j\Big), \quad (5)$$

since $\mathrm{P}_{x_i}\, x_j = \sin(\theta_j - \theta_i)\, Rx_i$ (Section A.4). AKOrN (Miyato et al., 2025) generalizes this to oscillators $x_i \in \mathbb{S}^{N-1}$, typically with $N \in \{2, 4\}$:

$$\dot x_i = \Omega_i x_i + \mathrm{P}_{x_i}\Big(c_i + \sum_j J_{ij} x_j\Big), \quad (6)$$

with antisymmetric natural frequencies Ω_i, couplings J_{ij} (a convolution or attention) and a conditional stimulus c_i. We call $c_i + \sum_j J_{ij} x_j$ the *drive*. The discrete update with learned step size $\gamma > 0$ is

$$x_i \leftarrow \Pi\Big(x_i + \gamma\Big[\Omega_i x_i + \mathrm{P}_{x_i}\Big(c_i + \sum_j J_{ij} x_j\Big)\Big]\Big). \quad (7)$$

For $\Omega = 0$ and symmetric couplings $J_{ij} = J_{ji}^\top$,

$$E(x) = -\tfrac{1}{2}\textstyle\sum_{i,j} x_i^\top J_{ij} x_j - \sum_i c_i^\top x_i \quad (8)$$

is a Lyapunov function of (6). A network stacks L blocks of T steps, passing a norm-based readout of each block as the next stimulus; it is trained by backpropagation through time, and on Sudoku clue cells start at $x = c$, the remaining cells start randomly, and test-time predictions are selected by energy-based voting over random initializations.

3.3. SFPRM

SFPRM (Anonymous, 2026) keeps the state on $\mathcal{S} := (\mathbb{S}^{d-1})^T$; the input $x \in \mathcal{S}$ is the token-wise normalized embedding. One iteration reads

$$\tilde z_i = F_\theta(z_i; x) := f_\theta\big(\mathrm{Slerp}_\lambda(z_i, x)\big), \quad (9)$$

$$z_{i+1} = \mathrm{Slerp}_{\eta_i}(z_i, \tilde z_i), \quad (10)$$

(geodesic injection and geodesic damping), and the solver halts when $\max_t 1 - \langle z_{i,t},\, \tilde z_{i,t}\rangle < \tau_c$. The damping η_i starts at 1 and is decayed by a patience rule. The weight-tied network f_θ is a stack of D nGPT layers, each consisting of

$$h \leftarrow \mathrm{Norm}\big(h + \alpha_A \odot (\mathrm{Norm}(\mathrm{Attn}(h)) - h)\big), \quad (11)$$

$$h \leftarrow \mathrm{Norm}\big(h + \alpha_M \odot (\mathrm{Norm}(\mathrm{MLP}(h)) - h)\big), \quad (12)$$

with per-coordinate eigen learning rates $\alpha_A, \alpha_M \in \mathbb{R}^d$. Multi-head attention with QK-normalization has the form of an AKOrN coupling with state-dependent weights,

$$\begin{aligned} \mathrm{Attn}(h)_t &= \textstyle\sum_{s=1}^T J_{ts}(h)\, h_s, \\ J_{ts}(h) &:= \textstyle\sum_{k=1}^H a_{ts}^k(h)\, W_O^k W_V^k, \end{aligned} \quad (13)$$

where $a_{ts}^k(h)$ are the softmax weights of head k.

4. Every Spherical Update Is a Tangent Step

Lemma 4.1 (Renormalized residual updates)**.** *Let $z \in \mathbb{S}^{n-1}$ and $w \in \mathbb{R}^n$ with $1 + \langle w,\, z\rangle > 0$. Then*

$$\Pi(z + w) = \Pi\Big(z + \frac{1}{1 + \langle w,\, z\rangle}\, \mathrm{P}_z\, w\Big). \quad (14)$$

Proof. By (1), $z + w = (1 + \langle w,\, z\rangle)\, z + \mathrm{P}_z\, w$, and $\Pi(\beta y) = \Pi(y)$ for $\beta > 0$. □

Lemma 4.2 (Slerp is a tangent step)**.** *Let $u, v \in \mathbb{S}^{n-1}$, $\theta = \angle(u, v) \in (0, \pi)$ and $\eta \in [0, 1]$. Then* $\mathrm{Slerp}_\eta(u, v) = \mathrm{Exp}_u(\eta\, \mathrm{Log}_u v)$*, and if $\eta\theta < \pi/2$,*

$$\mathrm{Slerp}_\eta(u, v) = \Pi\big(u + \gamma_\eta(\theta)\, \mathrm{P}_u\, v\big), \quad \gamma_\eta(\theta) := \frac{\tan(\eta\theta)}{\sin\theta}. \quad (15)$$

As $\theta \to 0$, $\gamma_\eta(\theta) \to \eta$; at $\eta = 1$, $\gamma_1(\theta) = 1/\cos\theta$.

Proof. Let $e := \mathrm{P}_u\, v/\sin\theta$, a unit tangent vector, so $v = \cos\theta\, u + \sin\theta\, e$. In (4) the coefficient of u is $[\sin((1-\eta)\theta) +$

$\sin(\eta\theta)\cos\theta]/\sin\theta = \cos(\eta\theta)$ and that of e is $\sin(\eta\theta)$, so $\mathrm{Slerp}_\eta(u, v) = \cos(\eta\theta)u + \sin(\eta\theta)e = \mathrm{Exp}_u(\eta\,\mathrm{Log}_u v)$. If $\eta\theta < \pi/2$ then $\cos(\eta\theta) > 0$, and dividing by it gives $\Pi(u + \tan(\eta\theta)\, e) = \Pi(u + \gamma_\eta(\theta)\,\mathrm{P}_u\, v)$. □

Remark 4.3 (Only the angle differs). All these updates move the state along the great circle through z in a tangent direction and differ only in the angle: $\Pi(z + \gamma\,\mathrm{P}_z\, h)$ rotates by $\arctan(\gamma\|\mathrm{P}_z\, h\|) < \pi/2$, $\mathrm{Slerp}_\eta(z, v)$ by $\eta\angle(z, v)$, and $\Pi(z + w)$ by $\mathrm{atan2}(\|\mathrm{P}_z\, w\|, 1 + \langle w,\, z\rangle)$. The positivity conditions exclude exactly the rotations by $\pi/2$ or more, which the AKOrN form cannot produce with $\gamma > 0$.

5. SFPRM Is an Operator-Split AKOrN

Definition 5.1 (Generalized AKOrN step)**.** Given a step size $\gamma > 0$ (possibly state-dependent), an antisymmetric Ω and a drive $D : \mathcal{S} \to (\mathbb{R}^d)^T$, define token-wise on $\mathcal{S}$

$$\mathcal{A}_{\gamma,\Omega,D}(z) := \Pi\big(z + \gamma\,[\,\Omega z + \mathrm{P}_z\, D(z)\,]\big). \tag{16}$$

AKOrN's update (7) is the case $D(x) = c + Jx$ with constant γ, applied to oscillators of dimension N.

5.1. One Iteration Is Exactly One AKOrN Step

Theorem 5.2 (Exact outer equivalence)**.** *Let $\theta_{i,t} := \angle(z_{i,t}, \tilde{z}_{i,t})$ and assume $\eta_i\theta_{i,t} < \pi/2$ for all t. Then the SFPRM iteration (9)–(10) satisfies*

$$z_{i+1,t} = \Pi\big(z_{i,t} + \gamma_{i,t}\ \mathrm{P}_{z_{i,t}}\, F_\theta(z_i; x)_t\big), \tag{17}$$

with $\gamma_{i,t} = \tan(\eta_i\theta_{i,t})/\sin\theta_{i,t}$ ($\gamma_{i,t} := \eta_i$ if $\theta_{i,t} = 0$). That is, $z_{i+1} = \mathcal{A}_{\gamma_i,0,F_\theta(\cdot;x)}(z_i)$: an AKOrN step with $\Omega = 0$, one oscillator per token ($N = d$), drive $c + Jz := F_\theta(z; x)$, and an angle-dependent step size.

Proof. Apply Lemma 4.2 token-wise to (10) with $u = z_{i,t}$, $v = \tilde{z}_{i,t}$. □

Corollary 5.3 (Fixed points)**.** *For any positive step size, the fixed points of $\mathcal{A}_{\gamma,0,F_\theta}$ are the states with $F_\theta(z; x)_t = \pm z_t$ for all t. The fixed points of SFPRM are those with $F_\theta(z; x) = z$; the antipodal solutions are excluded because* Slerp *is undefined there. The step size (SFPRM's η_i or AKOrN's γ) never changes the solution set.*

5.2. Inside the Iteration, Every Operation Is an AKOrN Step

Proposition 5.4 (Geodesic injection is the stimulus step)**.** *Let $\theta_t^x := \angle(z_t, x_t)$ and assume $\lambda\theta_t^x < \pi/2$ (automatic for $\lambda \le 1/2$). Then $\mathrm{Slerp}_\lambda(z, x)_t = \Pi\big(z_t + \gamma_\lambda(\theta_t^x)\ \mathrm{P}_{z_t}\, x_t\big)$: an AKOrN step with stimulus $c = x$, no coupling, and $\Omega = 0$.*

Proof. Lemma 4.2 with $u = z_t$, $v = x_t$, $\eta = \lambda$. □

Proposition 5.5 (Injection is a Riemannian gradient step)**.** *For $\theta_t^x \in (0, \pi)$, $\mathrm{Slerp}_\lambda(z, x)_t = \mathrm{Exp}_{z_t}\big(-\lambda\,\mathrm{grad}_{z_t}\,\Phi_x\big)$ with $\Phi_x(z) := \frac{1}{2}\sum_t \angle(z_t, x_t)^2$.*

Proof. On the sphere, $\mathrm{grad}_z\,\frac{1}{2}\angle(z, x)^2 = -\,\mathrm{Log}_z\, x$ for $\angle(z, x) \in (0, \pi)$; combine with Lemma 4.2. □

Thus SFPRM's stimulus is the gradient of a geodesic-distance potential, whereas AKOrN's stimulus term $\mathrm{P}_x\, c$ is the Riemannian gradient of the linear potential $-\langle c,\, x\rangle$ in (8). The anchor angle $\angle(z_t, x_t)$ analysed by Anonymous (2026) is the square root of twice this potential, token by token.

Proposition 5.6 (nGPT attention sub-layer is the coupling step)**.** *Let $A_t := \mathrm{Attn}(h)_t \neq 0$, $\hat{A}_t := A_t/\|A_t\|$, and $\alpha_A = a\mathbf{1}$ with $1 + a(\langle\hat{A}_t,\, h_t\rangle - 1) > 0$ (automatic for $a < 1/2$). Then (11) equals*

$$h_t \leftarrow \Pi\Big(h_t + \gamma_t^A\ \mathrm{P}_{h_t} \sum_s J_{ts}(h)\, h_s\Big), \tag{18}$$

with $\gamma_t^A = a\,/\,\big(\|A_t\|\,(1 + a(\langle\hat{A}_t,\, h_t\rangle - 1))\big)$: an AKOrN step with attention coupling, $c = 0$ and $\Omega = 0$.

For per-coordinate $\alpha_A \in \mathbb{R}^d$ the same holds with a diagonally preconditioned coupling and an additional self-drive $-\,\mathrm{diag}(\alpha_A)h_t$, whose tangent part is $-\,\mathrm{grad}\,\frac{1}{2}h^\top\,\mathrm{diag}(\alpha_A)h$ and vanishes when α_A is constant (Section B). The MLP sub-layer (12) is the same step with $\sum_s J_{ts}h_s$ replaced by the token-wise nonlinear drive $\mathrm{MLP}(h)_t$ (Proposition B.2). AKOrN has no counterpart inside a Kuramoto layer; its nonlinearity sits in the readout between blocks, which produces the next stimulus, so the MLP sub-layer acts as a state-dependent stimulus.

Theorem 5.7 (SFPRM iteration as operator-split AKOrN)**.** *Under the positivity conditions of Theorem 5.2 and Propositions 5.4 and 5.6, one SFPRM iteration is the composition*

$$\tilde{z}_i = \big(\mathcal{M}_D \circ \mathcal{C}_D \circ \cdots \circ \mathcal{M}_1 \circ \mathcal{C}_1 \circ \mathcal{I}\big)(z_i), \tag{19}$$

followed by $z_{i+1} = \mathcal{A}_{\gamma_i,0,F_\theta(\cdot;x)}(z_i)$, where $\mathcal{I}$ is the stimulus step (Proposition 5.4), $\mathcal{C}_\ell$ the coupling step of layer ℓ (Proposition 5.6) and $\mathcal{M}_\ell$ its self-drive step. Each factor is a generalized AKOrN step with $\Omega = 0$ and $N = d$.

Proof. Combine Propositions B.2, 5.4 and 5.6 for F_θ and Theorem 5.2 for the damping step. □

AKOrN instead evaluates stimulus and coupling in one Euler step with the summed drive $c + Jx$. The next result shows that the two agree to first order.

5.3. Collapse to a Single AKOrN Euler Step

Theorem 5.8 (First-order collapse)**.** *Use scalar eigen learning rates and let $\lambda = \varepsilon\kappa_0$, $\alpha_A^\ell = \varepsilon\kappa_A^\ell$, $\alpha_M^\ell = \varepsilon\kappa_M^\ell$, and fix $\eta \in (0, 1]$. Let $z_t \neq \pm x_t$ and all attention and MLP outputs be nonzero. Then, as $\varepsilon \to 0$,*

$$z_{i+1} = \Pi\big(z + \eta\varepsilon\ \mathrm{P}_z\, V(z)\big) + O(\varepsilon^2), \tag{20}$$

where $V_t(z) = c_t + (Jz)_t + m_t(z)$ with stimulus, coupling and self-drive

$$c_t = \kappa_0 \tfrac{\theta_t^x}{\sin\theta_t^x}\, x_t, \tag{21}$$

$$(Jz)_t = \textstyle\sum_{\ell=1}^D \frac{\kappa_A^\ell}{\|\mathrm{Attn}^\ell(z)_t\|} \sum_s J_{ts}^\ell(z)\, z_s, \tag{22}$$

$$m_t(z) = \textstyle\sum_{\ell=1}^D \kappa_M^\ell\ \mathrm{MLP}^\ell(z)_t / \|\mathrm{MLP}^\ell(z)_t\|. \tag{23}$$

This is AKOrN's update (7) with $\gamma = \eta\varepsilon$, $\Omega = 0$, $N = d$, stimulus c, attention coupling J, and an additional nonlinear self-drive m.

The proof (Section B) is a Lie–Trotter argument (Trotter, 1959): every intermediate state is $z + O(\varepsilon)$, so evaluating all drives at z costs $O(\varepsilon^2)$.

Remark 5.9 (Scope). Trained SFPRM models do not operate in the small-step regime: the reported injection fractions are $\lambda \approx 0.47$–0.58 (or λ fixed at 0.1 on Sudoku), and η_i starts at 1 (Anonymous, 2026). Theorem 5.8 is therefore structural: it identifies *which* AKOrN terms SFPRM contains. Theorems 5.2 and 5.7 are exact for any step sizes satisfying the positivity conditions.

5.4. Linearization and the Damping Window

Proposition 5.10 (Jacobian at a fixed point)**.** *Let $z^\star$ be a fixed point of $\mathcal{A}_{\gamma,\Omega,D}$ with D differentiable and γ smooth and positive near $z^\star$. Let $\mu_t := \langle D(z^\star)_t, z_t^\star\rangle$, J_D the Euclidean Jacobian of D at $z^\star$, P the token-wise projection at $z^\star$, and $\mathrm{M}_\mu := \mathrm{diag}(\mu_1 \mathrm{I}_d, \ldots, \mu_T \mathrm{I}_d)$. On $\mathrm{T}_{z^\star}\mathcal{S}$,*

$$\mathrm{d}\mathcal{A}_{\gamma,\Omega,D}(z^\star) = \mathrm{I} + \gamma\big(\mathrm{P}\, J_D\, \mathrm{P} - \mathrm{M}_\mu + \mathrm{P}\,\Omega\,\mathrm{P}\big). \tag{24}$$

Corollary 5.11 (SFPRM's damping window is the AKOrN window)**.** *For SFPRM, $\Omega = 0$, $D = F_\theta$ and $F_\theta(z^\star; x) = z^\star$, so $\mu_t = 1$ and $\mathrm{P}\, J_F\, \mathrm{P}$ restricted to the tangent space is the tangent Jacobian J of F_θ. Since $\gamma_\eta(\theta) \to \eta$ as $\theta \to 0$, both the* Slerp *step and its AKOrN form (Theorem 5.2) linearize to $(1-\eta)\mathrm{I} + \eta J$. The window*

$$\eta < \min_{\zeta \in \sigma(J)} 2(1 - \mathrm{Re}\,\zeta)/|\zeta - 1|^2 \tag{25}$$

of Anonymous (2026) therefore applies verbatim to the AKOrN form with γ in place of η.

Since $\mathrm{P}\,\Omega\,\mathrm{P}$ is skew-symmetric on the tangent space, a natural frequency adds a rotational component to $\mathrm{d}\mathcal{A}$ and pushes eigenvalues off the real axis. Modes with $\zeta - 1$ close to imaginary narrow the damping window and slow down halting (Anonymous, 2026), so any oscillatory extension must make the rotation end in locking (Section 7).

6. Where the Models Differ

The equivalence holds at the level of the update rule; Table 2 lists the remaining differences. The most consequential is the energy. In the small-step regime of Theorem 5.8 the drive is a Riemannian gradient field exactly when its coupling part is: the stimulus is $-\,\mathrm{grad}\,\Phi_x$ by Proposition 5.5, so

$$E(z) = \kappa_0 \Phi_x(z) - \tfrac{1}{2} \textstyle\sum_{t,s} z_t^\top \bar{J}_{ts}\, z_s \tag{26}$$

would be a Lyapunov function for a symmetric, state-independent $\bar{J}$ and no self-drive. Softmax attention is neither symmetric nor state-independent, so AKOrN's energy does not carry over; an energy-based attention block is required to recover it, which is what Section 8 builds.

SFPRM's theory in oscillator terms. (i) The obstruction to fixed-point existence cited by Anonymous (2026) ("rotating a circle moves every point") is the $\Omega \neq 0$ case; injection, the stimulus step of Proposition 5.4, pins the oscillators and restores existence. (ii) Global convergence under strong injection is the strong-stimulus regime of a driven oscillator network, in which all oscillators lock to a unique state. (iii) The anchor angle $\angle(z_t, x_t)$ is the phase lag of oscillator t relative to its stimulus.

7. An Oscillatory Extension of SFPRM

The correspondence suggests a family that contains SFPRM as the case $\Omega = 0$, $N = d$.

Definition 7.1 (Oscillatory SFPRM)**.** Partition each token into $M = d/N$ groups, each an oscillator on $\mathbb{S}^{N-1}$, and let Ω be antisymmetric and block diagonal, with frequencies $\omega_{m,k}$ for the $N/2$ planes of oscillator m. Replace (10) group-wise by

$$z_{i+1} = \mathrm{Exp}_{z_i}\big(\eta_i\big[\Omega z_i + \mathrm{Log}_{z_i}\,\Pi(\tilde{z}_i)\big]\big), \tag{27}$$

where $\tilde{z}_i = F_\theta(z_i; x)$ and $\Pi(\tilde{z}_i)$ normalizes each group of the proposal. By Lemma 4.2, $\Omega = 0$ and $N = d$ recover (10).

The fixed points satisfy $\Omega z + \mathrm{Log}_z\,\Pi(\tilde{z}) = 0$ group-wise: the proposal must sit at an angle $\|\Omega z\|$ behind the iterate (phase-lag locking). This condition does not depend on η_i, so the patience rule remains valid. At a locked state $1 - \langle z, \tilde{z}\rangle = 1 - \cos\|\Omega z\| \neq 0$, so the chordal residual never fires; the halting residual should be the tangent speed $\max_{t,m}\|\Omega z + \mathrm{Log}_z\,\Pi(\tilde{z})\|$.

Table 2. Differences between AKOrN and SFPRM beyond the shared update geometry.

	AKOrN	SFPRM
Oscillator size	$N \in \{2, 4\}$, many per position	$N = d$, one per token
Natural frequency	$\Omega \neq 0$ (learned global ω)	$\Omega = 0$
Drive	one coupling J plus stimulus c, summed in one Euler step	deep nGPT block; injection, coupling and self-drive as sequential steps; eigen-LR preconditioning
Stimulus	linear potential $-\langle c,\, x\rangle$, applied every step	geodesic potential $\frac{1}{2}\angle(z, x)^2$ (Proposition 5.5), once per iteration
Step size	constant learned γ	angle-dependent $\gamma_\eta(\theta)$; η decayed by patience
Energy	Lyapunov for symmetric J, $\Omega = 0$	none in general (F_θ is not a gradient field); restored by the energy model of Section 8
Outer structure	L blocks $\times$ T steps, readout between blocks	one block looped to a fixed point, chordal halting
Readout	$\|Wx\| + b$ per group, then MLP	linear $s_z h_\phi(z)$ with unnormalized rows
Initialization	clue cells at $x = c$, others random	random unit-norm
Training / inference	BPTT over T; larger T and energy voting at test time	deep supervision with truncated BPTT; damped solve until halting

Proposition 7.2 (A constant Ω is absorbed by the block)**.** *Fix the grouping and assume* $\|\Omega z + \mathrm{Log}_z\, \Pi(\tilde{z})\| < \pi$ *group-wise. With* $F'_\theta(z; x) := \mathrm{Exp}_z\big(\Omega z + \mathrm{Log}_z\, \Pi(F_\theta(z; x))\big)$*, the update* (27) *equals* $\mathrm{Slerp}_{\eta_i}\big(z_i, F'_\theta(z_i; x)\big)$ *for every* $\eta_i \in (0, 1]$.

Proof. $\mathrm{Log}_z \mathrm{Exp}_z(v) = v$ for tangent v with $\|v\| < \pi$, so $\mathrm{Log}_z F'_\theta = \Omega z + \mathrm{Log}_z\, \Pi(F_\theta)$; apply $\mathrm{Slerp}_\eta(z, v) = \mathrm{Exp}_z(\eta\, \mathrm{Log}_z\, v)$ from Lemma 4.2. □

A constant natural frequency is therefore a reparametrization of the block rather than a new solver. What cannot be absorbed is a change of the state space (the group size N) or anything that varies across iterations, such as an annealed frequency.

Proposition 7.3 (Locking condition for a frozen drive)**.** *Let* N *be even and* Ω *antisymmetric with rotation planes* $k = 1, \ldots, N/2$ *and frequencies* $\omega_k > 0$*. For a fixed drive* $h \in \mathbb{R}^N$ *with plane components* h_k*, the step* $x \mapsto \Pi(x + \gamma[\Omega x + \mathrm{P}_x\, h])$ *has a fixed point on* $\mathbb{S}^{N-1}$ *if and only if*

$$\sum_{k=1}^{N/2} \|h_k\|^2/\omega_k^2 \;\geq\; 1. \tag{28}$$

For $N = 2$ *this is the Adler condition* $\|h\| \geq \omega$ *(Adler, 1946).*

In a network the drive depends on the state, so (28) is a mean-field condition per oscillator. It nevertheless gives a reading of the halting rule: an oscillator is locked when its drive $\|h_k\|$ outweighs its frequency ω_k. Near the boundary of (28) the relaxation rate tends to zero (for $N = 2$ it equals $\sqrt{\|h\|^2 - \omega^2}$), so the number of iterations to halt grows. Truncated-BPTT gradients degrade there as well, which motivates a margin penalty on (28) at the final state. Whether the extension improves trained reasoners is untested.

8. An Energy-Based SFPRM

Section 6 identified the energy as the main structural difference between the two models: AKOrN's drive is a gradient field when J is symmetric, whereas SFPRM's F_θ never is. We now close this gap from the SFPRM side. The recurrent map is replaced by a Riemannian gradient step on an explicit energy $E(w; x)$, so that every iteration is an AKOrN step whose drive is $-\nabla E$. The injection is already a gradient (Proposition 5.5); for attention and the MLP we take the two energies of the Energy Transformer (Hoover et al., 2023) to the sphere, with modifications that reasoning tasks turn out to require (Section 8.2).

8.1. Energy, Update and Gradient

Definition 8.1 (Energy SFPRM)**.** The state is $w = (z, u) \in (\mathbb{S}^{d-1})^{2T}$ with visible tokens z and hidden tokens u, one per visible token; the *flat* variant identifies $u \equiv z$. With heads $k = 1, \ldots, H$, signed head weights $\sigma_k \in \mathbb{R}$, logit scales $\beta_k > 0$, memories $\xi_1, \ldots, \xi_K \in \mathbb{R}^d$ (rows of Ξ) and $\kappa_0, \kappa_C, \kappa_M, \beta_M > 0$,

$$\begin{aligned} E(w; x) &= \kappa_0\, \Phi_x(z) + E_A(u, z) + E_C(u, z) + E_M(u), \\ E_A &= -\textstyle\sum_{k=1}^{H} \frac{\sigma_k}{\beta_k} \sum_t \log \sum_{s \neq t} e^{\ell^k_{ts}}, \\ E_C &= -\kappa_C \textstyle\sum_t \langle u_t,\, C z_t\rangle, \\ E_M &= -\dfrac{\kappa_M}{\beta_M} \textstyle\sum_t \log \sum_{\mu=1}^{K} e^{\beta_M \langle \xi_\mu,\, u_t\rangle}, \end{aligned} \tag{29}$$

where $\ell^k_{ts} := \beta_k \langle W^k_Q u_t,\, W^k_K z_s\rangle$ are the attention logits and $\Phi_x(z) = \frac{1}{2}\sum_t \angle(z_t, x_t)^2$ is the injection potential of Proposition 5.5. The recurrent map is the unit gradient step $F_E(w; x) := \mathrm{Exp}_w\big(-\mathrm{grad}\, E(w; x)\big)$, with grad the Riemannian gradient on $(\mathbb{S}^{d-1})^{2T}$, and the iteration is SFPRM's damped loop (10) with F_E in place of F_θ:

$$\begin{aligned} w_{i+1} &= \mathrm{Slerp}_{\eta_i}\big(w_i, F_E(w_i; x)\big) \\ &= \mathrm{Exp}_{w_i}\big(-\eta_i\, \mathrm{grad}\, E(w_i; x)\big), \end{aligned} \tag{30}$$

the equality by Lemma 4.2 whenever $\|\mathrm{grad}_{w_t} E\| < \pi$. The prediction is read from z. SFPRM's chordal halting residual of F_E is $1 - \cos\|\mathrm{grad}_{w_t} E\|$, so the halting test is $\max_t \|\mathrm{grad}_{w_t} E\| < \tau_g$: the iteration halts at τ_g-stationary points of E.

Proposition 8.2 (The gradient is injection, symmetric coupling and memory)**.** *Let* $a^k_{ts} = \mathrm{softmax}_{s\neq t}\big(\beta_k \langle W^k_Q u_t,\, W^k_K z_s\rangle\big)$ *and* $M_k := (W^k_Q)^\top W^k_K$. *Token-wise,*

$$-\mathrm{grad}_{z_t} E = \kappa_0 \,\mathrm{Log}_{z_t} x_t + \kappa_C \,\mathrm{P}_{z_t} C^\top u_t + \mathrm{P}_{z_t} \textstyle\sum_k \sigma_k \sum_{s\neq t} a^k_{st} M_k^\top u_s, \tag{31}$$

$$-\mathrm{grad}_{u_t} E = \mathrm{P}_{u_t}\Big[\textstyle\sum_k \sigma_k \sum_{s\neq t} a^k_{ts} M_k z_s + \kappa_C C z_t + \kappa_M \Xi^\top \mathrm{softmax}(\beta_M \Xi u_t)\Big]. \tag{32}$$

In the flat variant the two attention sums add, and the drive is an AKOrN coupling $\sum_s J_{ts}(z) z_s$ *with* $J_{ts} = \sum_k \sigma_k\big(a^k_{ts} M_k + a^k_{st} M_k^\top\big)$*, which satisfies AKOrN's symmetry condition* $J_{ts} = J_{st}^\top$ *at every state.*

Every term has a counterpart in Theorem 5.8: $\kappa_0 \,\mathrm{Log}_z x$ is the stimulus c, the attention sums are the coupling Jz, and the memory is the self-drive m. The energy model is therefore the collapsed single-step form of SFPRM in which the coupling is a gradient field and the MLP is a Hopfield memory: the query-path term $\sum_s a_{ts} M_k z_s$ is standard attention with value matrix M_k (rank at most d_h), the key-path term $\sum_s a_{st} M_k^\top u_s$ is the contribution that makes the field conservative (Hoover et al., 2023), the matrix C plays the role of the output projection, and $\Xi^\top \mathrm{softmax}(\beta_M \Xi u)$ is the tied-weight MLP $W_2\, r(W_1 u)$ with $W_2 = W_1^\top$ and $r = \mathrm{softmax}$, the retrieval rule of modern Hopfield networks (Krotov & Hopfield, 2016; Ramsauer et al., 2021); $r = \mathrm{relu}$ with $G(a) = \frac{1}{2}\mathrm{relu}(a)^2$ in place of the log-sum-exp gives the classical memory. In the terms of Table 2, the energy row now reads "yes" for both models.

8.2. Three Design Choices

The Energy Transformer was designed for image completion and graph tasks. Three modifications, each forced by a failure we observed in the pilot of Section 8.5, make (29) suitable for reasoning while keeping it an energy. *Signed heads.* With $\sigma_k > 0$ the log-sum-exp is minimized by aligning queries with keys: attention is attractive. A head with $\sigma_k < 0$ is repulsive, and its drive pushes a token away from the tokens it attends to most. Constraints of the form "cells in the same unit hold different digits" are repulsive couplings, the negative couplings of the Kuramoto model (Kuramoto, 1984), and cannot be expressed with attractive heads alone. *Hidden tokens.* In the flat variant the message $M_k z_s$ delivered to a token is tied to the score $\langle W^k_Q z_t,\, W^k_K z_s\rangle$ that selects it, since both use M_k; a model cannot match on position and message on content. Hidden tokens receive the query-path message, the visible token receives $C^\top u_t$ through the free bilinear coupling E_C, and matching and messaging decouple. The energy becomes a two-layer associative memory (Krotov, 2021); the state doubles, the theory below is unchanged. *Sharpness versus message size.* In (29) the softmax sharpness is $\beta_k \|W^k_Q u_t\| \|W^k_K z_s\|$ while the message size is $|\sigma_k| \|M_k z_s\|$, so a small W with a large β_k gives sharp attention and unit-scale messages at the same time, the separation that QK-normalization and the logit scale achieve in nGPT. Sizing W for unit logits instead makes $\|M_k\|$ large, the Hessian of E_A grows with $\beta_k \|M_k\|^2$, and the iteration leaves the stability window of Proposition 8.4.

8.3. What the Energy Buys

Theorem 8.3 (Energy descent, global convergence and a halting bound)**.** *Let* $L > 0$ *be such that* $E(\mathrm{Exp}_w(v)) \le E(w) + \langle \mathrm{grad}\, E(w),\, v\rangle + \frac{L}{2}\|v\|^2$ *for all* w *and tangent* v*. Such an* L *exists;* $L = \sup_w \lambda_{\max}(\mathrm{Hess}\, E(w))$ *suffices, and the injection contributes at most* κ_0 *to it because the Hessian of* $\frac{1}{2}\angle(\cdot, x)^2$ *has eigenvalues* 1 *and* $\theta \cot\theta \le 1$*. Fix* $\eta \in (0, 2/L)$ *and let* $c_\eta := \eta(1 - \eta L/2) > 0$*. Then along every trajectory on which no token is antipodal to its input (an exceptional set of measure zero, empty for the chordal injection* $\sum_t (1 - \langle z_t,\, x_t\rangle)$*):*

(i) $E(w_{i+1}) \le E(w_i) - c_\eta \|\mathrm{grad}\, E(w_i)\|^2$*: E is a Lyapunov function;*

(ii) $\sum_{i\ge 0} \|\mathrm{grad}\, E(w_i)\|^2 \le \big(E(w_0) - \min E\big)/c_\eta$*, so* $\mathrm{grad}\, E(w_i) \to 0$*, every accumulation point is stationary, and the halting test* $\max_t \|\mathrm{grad}_{w_t} E\| < \tau_g$ *fires within* $\big\lceil (E(w_0) - \min E)/(c_\eta \tau_g^2) \big\rceil$ *iterations;*

(iii) if E is real-analytic on a neighbourhood of the accumulation set, the whole sequence converges to a single stationary point.

The same holds for decaying $\eta_i \in [\eta_{\min}, \eta]$ *with* c_η *replaced by* $\min(c_{\eta_{\min}}, c_\eta)$*, so SFPRM's patience rule is admissible.*

SFPRM has existence of a fixed point and convergence from starts near one (Anonymous, 2026). With an energy every start settles, and the number of iterations before the halting test fires is bounded a priori. The energy also replaces the patience heuristic: accept η_i if $E(w_{i+1}) \le E(w_i) - c\,\eta_i \|\mathrm{grad}\, E(w_i)\|^2$ and halve it otherwise (Armijo backtracking), which guarantees (i) at every iteration without knowledge of L. Analyticity in (iii) holds for the log-sum-exp terms, the bilinear term and, away from the antipode, the squared angle; the antipode is the point at which Slerp, and hence SFPRM's own update, is undefined (Corollary 5.3).

Proposition 8.4 (Real spectrum and the window at a halted state)**.** *Let $w^\star$ be a stationary point of E at which E is twice differentiable. On $\mathrm{T}_{w^\star}(\mathbb{S}^{d-1})^{2T}$,*

$$\begin{aligned} \mathrm{d}F_E(w^\star) &= \mathrm{I} - \mathrm{Hess}\, E(w^\star), \\ \mathrm{Hess}\, E(w^\star) &= \mathrm{P}\, \nabla^2 E\ \mathrm{P} - \mathrm{M}_\mu, \end{aligned} \tag{33}$$

with $\mu_t = \langle \nabla_{w_t} E(w^\star),\, w_t^\star \rangle$ as in Proposition 5.10, and the damped step (30) linearizes to $\mathrm{I} - \eta\, \mathrm{Hess}\, E(w^\star)$. The Hessian is self-adjoint, so every eigenvalue $\zeta = 1 - h$ of $\mathrm{d}F_E(w^\star)$ is real; the window of Corollary 5.11 is $\eta < 2/\lambda_{\max}(\mathrm{Hess}\, E(w^\star))$; $w^\star$ is attracting if and only if $\mathrm{Hess}\, E(w^\star) \succ 0$; and a strict saddle has an expanding mode $|\zeta| > 1$ for every $\eta > 0$.

For SFPRM, $\mathrm{d}F_\theta$ has no such structure, and Anonymous (2026) observe that modes with $\zeta - 1$ close to the imaginary axis narrow the window and slow down halting. The energy removes them: with real ζ the per-mode decay factor $|1-\eta h|$ is minimized by $\eta = 1/h$ and the window $2/h$ is the widest that Corollary 5.11 allows. Since the retracted gradient step avoids strict saddles from almost every start (Lee et al., 2019), halted states are local minima of E. nGPT's eigen learning rates survive as a fixed diagonal preconditioner $D \succ 0$ applied to the gradient: descent and real spectrum are preserved because $(\mathrm{P}\, D\, \mathrm{P})\, \mathrm{Hess}\, E$ is similar to a symmetric matrix (Section D).

Proposition 8.5 (Energy selection over restarts)**.** *Let $w^{(1)}, \ldots, w^{(B)}$ be halted states from B independent random starts. Every halted state has chordal residual below $1 - \cos \tau_g$, so the residual cannot rank them, whereas $E(w^{(b)}; x)$ can. Energy selection, $\hat{b} = \arg\min_b E(w^{(b)}; x)$, contains anchor-angle selection $\arg\min_b \Phi_x(z^{(b)})$ as the special case in which only the first term of (29) is kept. If the global minimizer of $E(\cdot; x)$ decodes to the correct answer, its basin of attraction has positive measure, and its energy is separated from every other stationary value by more than twice the halting slack, then the selected answer is correct with probability tending to one as $B \to \infty$.*

Energy-based voting is AKOrN's best test-time result: on the harder Sudoku test set, board accuracy rises from 51.7% without voting to 81.6% with 100 restarts and 89.5% in the configuration of their main table (Miyato et al., 2025), where majority voting does not help. SFPRM cannot do this today: every halted state has residual zero, and Proposition 8.5 says what it would need.

8.4. Training

Implicit gradients by conjugate gradient. At an attracting halted state the Hessian is positive definite, and the implicit function theorem gives, for a loss $\ell(w^\star)$,

$$\begin{aligned} \frac{\mathrm{d}\ell}{\mathrm{d}\theta} &= -\big\langle v,\ \partial_\theta\, \mathrm{grad}\, E(w^\star; \theta) \big\rangle, \\ \mathrm{Hess}\, E(w^\star)\, v &= \mathrm{grad}\, \ell(w^\star). \end{aligned} \tag{34}$$

The system is symmetric positive definite, so conjugate gradient (Hestenes & Stiefel, 1952) reaches relative error ϵ in $O(\sqrt{\kappa} \log(1/\epsilon))$ Hessian-vector products, each one backward pass, where κ is the condition number of $\mathrm{Hess}\, E(w^\star)$ (Nocedal & Wright, 2006). Truncated BPTT with window K, the training scheme of FPRM and SFPRM, sums the first K terms of the Neumann series $\eta \sum_k (\mathrm{I} - \eta\, \mathrm{Hess}\, E)^k$ for the same inverse (Section A), a Richardson iteration that needs $O(\kappa \log(1/\epsilon))$ terms. For SFPRM the matrix $\mathrm{I} - J$ is not symmetric and conjugate gradient does not apply.

Equilibrium propagation. Because (30) is gradient descent on E, the nudged energy $E + \beta\ell$ has a minimizer $w_\beta^\star$ near $w^\star$ and

$$\frac{\mathrm{d}\ell}{\mathrm{d}\theta} = \lim_{\beta \to 0} \beta^{-1} \big[\partial_\theta E(w_\beta^\star) - \partial_\theta E(w^\star) \big] \tag{35}$$

(Scellier & Bengio, 2017), with error $O(\beta)$, or $O(\beta^2)$ for a centered difference. The rule needs only the forward dynamics, run twice, and is the route by which natural frequencies of Kuramoto networks are learned in hardware (Ahmadi, 2026). SFPRM has no energy to nudge.

8.5. Expressivity and a First Pilot

The restriction is real: each head's value matrix is $M_k = (W_Q^k)^\top W_K^k$, every head carries a key-path term, the memory is a tied-weight MLP, and depth is replaced by hidden tokens. We compared the energy model of Definition 8.1 with SFPRM on 4×4 and 6×6 Sudoku, with the same embedding, readout, training recipe (deep supervision with truncated BPTT) and halting test, training on puzzles with many clues and testing in distribution (ID) and on puzzles with fewer clues (OOD); Section E has the protocol and the full results, and Table 3 lists them next to the variant of Section 8.6. The structural predictions hold: at every halted state the Jacobian is symmetric to machine precision and its spectrum is real, where 96% of SFPRM's eigenvalues are complex, and with Armijo backtracking the energy never increases. The model itself is weak. On 4×4 it reaches 82.4% ID and 17.2% OOD after six times more training steps than SFPRM (97.8% and 65.6%), and on 6×6 it solves no board within the budget in which SFPRM reaches 99.8% and 89.2%. The learned energy is stiff ($\lambda_{\max}(\mathrm{Hess}\, E) \approx$ 12–18, window ≈ 0.15), so halting is slower, not faster: a median of 68 OOD iterations against 19. Energy voting fails as well: the lowest-energy state among 64 restarts lowers 4×4 OOD accuracy to 12.4%, because the lowest minima of the learned energy are not the

Table 3. Energy SFPRM against SFPRM on 4×4 and 6×6 Sudoku (500 test puzzles per set; board accuracy in %, OOD unless stated; protocol in Section E). "Energy": the model of Definition 8.1 (Section 8.5). "Codebook": the variant of Section 8.6 with answer codebook, tied readout and embedding and the β schedule ($\rho = 8$, $K = 8$); "+rank" adds the ranking loss ($m = 1$, $R = 2$). "Mem. MLP": SFPRM with only its MLP replaced by the tied memory $\Xi^\top \operatorname{softmax}(\beta \Xi h)$, no energy. Energy models halt with Armijo backtracking, SFPRM models with the patience rule. Iterations to halt over halted puzzles; spectrum of $\mathrm{d}F$ at polished halted states; B=64: accuracy of the restart chosen by each rule among 64; committed: boards whose tokens are all committed ($\epsilon = 0.1$) at the halted state.

	4×4					6×6			
	SFPRM	Energy	Codebook	+rank	Mem. MLP	SFPRM	Energy	+rank	Mem. MLP
Parameters ($\times 10^3$)	101	79	79	79	67	101	80	80	68
Steps to 90% train acc.	200	1250	650	750	200	400	> 4000	> 4000	500
Board acc. ID	97.8	82.4	98.0	94.6	95.8	99.8	0.0	61.2	99.2
Board acc. OOD	65.6	17.2	74.4	62.8	55.0	89.2	0.0	11.0	69.8
Halted (≤ 128 it.)	73.4	81.8	98.0	97.4	75.4	90.6	30.4	25.0	77.6
Iterations (med./p90)	19/45	68/114	43/76	49/83	25/48	19/37	102/122	103/124	22/45
Complex ζ (%)	96.4	0.0	0.0	0.0	94.9	96.4	0.0	0.0	94.8
$\lambda_{\max}(\mathrm{Hess}\, E)$	–	12.2	12.5	12.7	–	–	17.9	12.9	–
Window (Cor. 5.11)	1.40	0.16	0.16	0.15	1.24	1.33	0.12	0.16	1.24
B=64: lowest energy	–	12.4	99.0	100.0	–	–	0.0	88.2	–
B=64: anchor angle	48.8	20.4	25.2	14.8	48.4	92.8	0.0	0.4	63.4
B=64: majority	67.0	20.6	91.2	84.8	56.4	90.6	0.0	59.4	71.4
B=64: oracle	81.8	36.0	99.0	100.0	67.0	93.8	0.0	89.8	82.2
Committed boards	–	–	98.4	95.8	–	–	–	4.6	–
Acc. committed/uncomm.	–	–	75.6/0.0	65.3/4.8	–	–	–	78.3/7.8	–

correct boards and deep supervision never asks them to be. The next subsection changes the memory so that they are.

8.6. The Memory as a Codebook: Readout, Commitment and Annealing

The memory term has a probabilistic reading that suggests three further changes. With memories of fixed norm $\|\xi_\mu\| = r$, $e^{\beta\langle \xi_\mu, z\rangle}$ is, up to normalization, a von Mises–Fisher density on $\mathbb{S}^{d-1}$ with mean direction ξ_μ / r and concentration βr (Mardia & Jupp, 2000), so $E_M(z_t) = -\frac{1}{\beta} \log p(z_t) + \mathrm{const}$ for an equal-weight vMF mixture centred on the memories, and the memory step is Riemannian gradient ascent on a learned mixture density. Each term of (29) then has a role in a continuous relaxation of a discrete constraint problem: the injection is the evidence, the attention energy holds the pairwise constraints, and the memory is a unary potential whose minima are the codebook points. At large β the memory energy of a token approaches $-\max_\mu \langle \xi_\mu, z_t\rangle$, a pinning potential toward the nearest prototype, the learned analogue of the sub-harmonic injection locking that oscillator Ising machines use to force continuous phases onto discrete values (Wang & Roychowdhury, 2019). This motivates the following *codebook variant* of Definition 8.1. *Answer codebook and tied readout.* An extra memory term $E_{\mathrm{ans}}(z) = -\frac{\kappa_{\mathrm{ans}}}{\beta_{\mathrm{ans}}} \sum_t \log \sum_{k=1}^n e^{\beta_{\mathrm{ans}} \langle \xi_k^{\mathrm{ans}}, z_t\rangle}$ on the visible tokens has one unit-norm prototype per output symbol; the readout is $\mathrm{logits}_t = s\, \Xi^{\mathrm{ans}} z_t$, so the prediction is the prototype the token sits in, the cross-entropy trains the prototypes directly, and the digit embedding of the input shares the same prototypes. *Commitment.* $c_t = \max_k \operatorname{softmax}(\beta_{\mathrm{ans}} \Xi^{\mathrm{ans}} z_t)_k$ is a label-free certificate; a token is committed when $c_t > 1 - \epsilon$, and "residual below τ and every token committed" is a stronger halting test than the residual alone. *Annealing.* β controls the landscape (Ramsauer et al., 2021): small β gives a single smooth minimum near the mean of the memories, large β one minimum per memory. A schedule $\beta_i = \beta\, \rho^{-(1-\min(i/K,1))}$ over iterations lets coupling and injection act first and commits tokens late; like an annealed frequency (Section 7) it cannot be folded into the weight-tied block, and Theorem 8.3 applies once the schedule has ended. *Ranking loss.* To make the premise of Proposition 8.5 hold, training adds, over R restarts of the same puzzle, the hinge $\max(0, E(w^\star_{\mathrm{correct}}) - E(w^\star_{\mathrm{wrong}}) + m)$ whenever both kinds of halted state occur, so that the energy learns to rank solutions. Table 3 repeats the pilot protocol for the codebook variant and Figure 1 shows its 4×4 spectrum and restart curves; Section E has the training and halting curves. Five findings. *The codebook changes the picture.* Without the ranking loss the variant solves 98.0% of ID and 74.4% of OOD boards on 4×4 with one restart, against 97.8% and 65.6% for SFPRM and 82.4% and 17.2% for the first energy model, and it trains in fewer steps than the first energy model. *Energy selection now works.* Selecting the lowest-energy halted state among B restarts matches the oracle at every B: OOD accuracy rises from 75.4% at $B = 1$ to 99.0% at $B = 64$, whereas majority voting reaches 91.2% and anchor-angle selection falls to 25.2%. SFPRM's best rule at $B = 64$ reaches 67.0% and its oracle 81.8% (Fig-

Table 4. Numerical verification of the identities in double precision on random instances. Errors are maximum absolute errors.

Check	Result
Lemma 4.1 (scalar and per-coordinate α)	1.7×10^{-16}
Lemma 4.2: $\mathrm{Slerp} = \mathrm{Exp} \circ \mathrm{Log} = \Pi(u + \gamma_\eta \, \mathrm{P}_u \, v)$	2.2×10^{-16}
Proposition 5.6: attention sub-layer	1.1×10^{-16}
Proposition 7.3: locked iff (28)	locks at 1.01, drifts at 0.99
Theorem 5.8 at $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}$	1.9×10^{-2}, 1.6×10^{-4}, 1.6×10^{-6}
Proposition 5.10, $\Omega = 0$ / $\Omega \neq 0$	7.6×10^{-11} / 8.7×10^{-11}
Proposition 8.2: $-\operatorname{grad} E$ in closed form, flat / hidden	4.4×10^{-16} / 4.4×10^{-16}
Proposition 8.2: $J_{ts} = J_{st}^\top$	0
Proposition 8.4: $\mathrm{d}F_E = \mathrm{I} - \mathrm{Hess}\, E$; asymmetry; $\max \lvert \mathrm{Im}\, \zeta \rvert$	4.0×10^{-16}; 8.2×10^{-17}; 0
Proposition 8.4: vs. finite differences	7.6×10^{-11}
Random SFPRM map at a fixed point: asymmetry; complex ζ	0.20; 84%
Theorem 8.3: increases of E at $\eta \leq 1.9/\hat{L}$ / $3/\hat{L}$	none in 4,000 steps / 54%
(34): adjoint vs. finite differences	1.7×10^{-9}
(34): CG / Neumann steps to 10^{-6}	11 / 81
(35): centered EP error at $\beta = 10^{-3}$	1.1×10^{-5}

ure 1). The premise of Proposition 8.5 holds for the learned energy: whenever a restart finds the solution, it is the lowest minimum. *The schedule is necessary, the ranking loss is not.* With β fixed at its final value from the first iteration the variant never leaves the plateau (no board solved, no puzzle halted, in 2,000 steps): tokens commit to prototypes before attention can propagate the constraints. The ranking loss does not improve selection, which already matches the oracle without it, and it costs single-restart accuracy (94.6% / 62.8% with it, although it places the correct state lower in energy for 88% of training pairs); energy selection behaves well because the readout and the energy share the prototypes, not because ranking is trained explicitly. *Commitment certifies.* At the halted state, 75.6% of committed OOD boards are correct and none of the uncommitted ones. Requiring commitment in the halting test does not help, however: it lowers board accuracy by 0.4–3.0 points, since almost every state that passes the residual test is already committed. *Spectrum and halting.* All eigenvalues at halted states are real, as Proposition 8.4 requires, and Armijo backtracking halts 98% of the OOD puzzles, against 24% for the patience rule. The learned Hessian remains stiff ($\lambda_{\max} \approx 13$), so halting takes a median of 43 OOD iterations against 19 for SFPRM; the β schedule does not remove the stiffness. Replacing only SFPRM's MLP by a tied memory, without any energy, costs a third of the parameters and some accuracy (95.8% / 55.0% on 4×4, 99.2% / 69.8% on 6×6) and leaves 94%

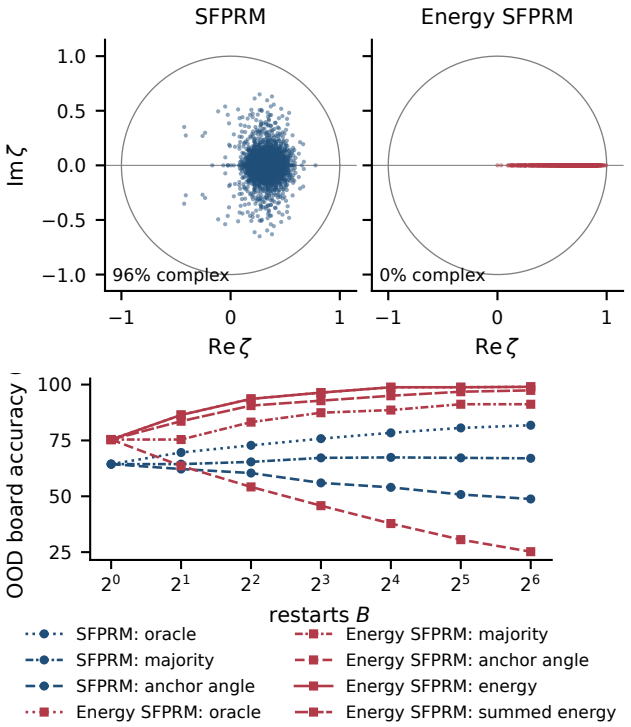


Figure 1. Codebook variant against SFPRM on 4×4 Sudoku. Top: eigenvalues of the tangent Jacobian at polished halted OOD states (three puzzles pooled); SFPRM at $\eta = 1$, the energy model for the damped map at $\eta = 1/\lambda_{\max}(\mathrm{Hess}\, E)$, half its window. Bottom: OOD board accuracy against the number of restarts B under the selection rules of Proposition 8.5; lowest-energy selection coincides with the oracle.

of the eigenvalues complex: the benefits above come from the energy, not from the memory layer alone. 6×6. The first energy model solved no 6×6 board; the codebook variant (trained with the ranking loss) reaches 61.2% ID and 11.0% OOD with one restart, below SFPRM's 99.8% and 89.2%, and its training accuracy was still rising when the 4,000-step budget ended. Energy selection again tracks the oracle: over 64 restarts it reaches 100% ID and 88.2% OOD (oracle 100% and 89.8%), against 59.4% for majority voting and 0.4% for the anchor angle OOD, and just below SFPRM's best rule at B=64 (92.8%). The spectrum is again real, halting is again slow (median 103 OOD iterations), and 98% of committed ID boards are correct against 52% of uncommitted ones.

9. Numerical Verification

We checked every identity in double precision on random instances with unit states and inputs, random weights, and multi-head attention with QK-normalization for Proposition 5.6 (Table 4; setup in Section C). The exact identities hold to machine precision. The error of Theorem 5.8 falls by a factor of 100 for every factor-10 decrease of ε,

as predicted for $O(\varepsilon^2)$. The Jacobian of Proposition 5.10 matches central finite differences to 10^{-10}, the accuracy of the differencing. The energy statements of Section 8 were checked in the same way (Section F). The closed-form gradient of Proposition 8.2 matches automatic differentiation of (29), the coupling is exactly symmetric, and at a stationary point the Jacobian of F_E equals $\mathrm{I} - \mathrm{Hess}\, E$ with a symmetric Hessian and a purely real spectrum, whereas a random SFPRM map at a fixed point has a Jacobian with relative asymmetry 0.20 and 84% of its eigenvalues off the real axis. The energy never increased in $4{,}000$ gradient steps at $\eta \in \{0.5, 1, 1.9\}/\hat{L}$, with $\hat{L}$ the largest Hessian eigenvalue found along descent trajectories, the descent inequality of Theorem 8.3(i) held at every step, and at $\eta = 3/\hat{L}$, outside the window, 54% of the steps increased the energy; the implicit gradient (34) matches finite differences, conjugate gradient needs 11 Hessian-vector products where the Neumann series of truncated BPTT needs 81 at condition number 6.5, and the equilibrium-propagation estimate (35) converges with the predicted $O(\beta^2)$ error.

10. Discussion and Limitations

The correspondence places SFPRM inside the oscillator family as its zero-frequency, single-oscillator, operator-split member, and places AKOrN's Euler step inside SFPRM's analysis: the existence result, the damping window and the halting test transfer to the AKOrN form, while the energy and the locking picture describe what SFPRM's injection does. Several limitations apply. First, the equivalence is a statement about update rules; readouts, initialization and training differ (Table 2), so it makes no claim that the two models learn the same functions. Second, the single-step collapse holds only for small step sizes, which trained SFPRM models do not use (Remark 5.9). Third, the oscillatory extension and the locking-margin penalty are proposals: we report no experiments on trained models, and Proposition 7.2 shows that a constant frequency alone cannot add expressivity. Fourth, the verification uses random instances rather than trained weights. Fifth, the energy-based SFPRM trades expressivity for guarantees: its attention is a conservative field with tied value matrices and its MLP a tied-weight memory. The pilot studies of Sections 8.5 and 8.6 are small and single-seeded, on small Sudoku grids where test grids can coincide with training grids; they show that the codebook variant can match SFPRM and that its energy can rank solutions, but not that this holds at scale. The learned energies are stiff, so halting remains slower than in SFPRM, and the β schedule voids the descent guarantee during its first iterations. Testing the non-absorbable oscillatory variants and the energy model on reasoning benchmarks at scale, with conjugate-gradient implicit training and energy voting, is the natural next step.

11. Conclusion

We showed that every update in SFPRM is an AKOrN Euler step with zero natural frequency: exactly at the level of one iteration, exactly for injection and attention inside it, and to first order for the iteration as a single step. The two forms share their fixed points and linearization, so SFPRM's damping window applies to the AKOrN form. The correspondence reads SFPRM's guarantees as oscillator locking, identifies the absence of an energy function as the main structural difference, and yields an oscillatory extension with an explicit locking condition, in which a constant frequency reduces to a reparametrization of the block. Closing the structural gap from the other side, an energy-based SFPRM makes the recurrent map a gradient step on an explicit energy and thereby gains a Lyapunov function, a halting bound, a real spectrum at halted states, energy-based voting and equilibrium propagation. With a codebook memory tied to the readout and an annealed concentration, the learned energy ranks solutions well enough that energy voting matches the oracle in our pilot.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

Absil, P.-A., Mahony, R., and Andrews, B. Convergence of the iterates of descent methods for analytic cost functions. *SIAM Journal on Optimization*, 16(2):531–547, 2005.

Absil, P.-A., Mahony, R., and Sepulchre, R. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, Princeton, NJ, 2008.

Adler, R. A study of locking phenomena in oscillators. *Proceedings of the IRE*, 34(6):351–357, 1946.

Ahmadi, M. R. The phase is the gradient: Equilibrium propagation for frequency learning in Kuramoto networks, 2026. arXiv:2604.10272.

Anonymous. Spherical fixed-point reasoning models, 2026. Under review.

Bai, S., Kolter, J. Z., and Koltun, V. Deep equilibrium models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Boumal, N. *An Introduction to Optimization on Smooth Manifolds*. Cambridge University Press, 2023.

Darlow, L., Regan, C., Risi, S., Seely, J., and Jones, L. Continuous thought machines, 2025. arXiv:2505.05522.

Daulbaev, T., Glazkov, I., Rakhuba, M., and Oseledets, I. Reasoning on the simplex: Geometric fixed-point models, 2026. arXiv:2609.33540.

Dehghani, M., Gouws, S., Vinyals, O., Uszkoreit, J., and Kaiser, Ł. Universal transformers. In *International Conference on Learning Representations (ICLR)*, 2019.

Hestenes, M. R. and Stiefel, E. Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards*, 49(6):409–436, 1952.

Hoover, B., Liang, Y., Pham, B., Panda, R., Strobelt, H., Chau, D. H., Zaki, M. J., and Krotov, D. Energy transformer. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Huang, B., Geng, Z., and Kolter, Z. Equilibrium reasoners: Learning attractors enables scalable reasoning, 2026. arXiv:2605.21488.

Jolicoeur-Martineau, A. Less is more: Recursive reasoning with tiny networks, 2025. arXiv:2510.04871.

Keller, T. A. and Welling, M. Neural wave machines: Learning spatiotemporally structured representations with locally coupled oscillatory recurrent neural networks. In *International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, 2023.

Krotov, D. Hierarchical associative memory. In *arXiv preprint arXiv:2107.06446*, 2021.

Krotov, D. and Hopfield, J. J. Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.

Kuramoto, Y. *Chemical Oscillations, Waves, and Turbulence*. Springer, Berlin, 1984.

Lee, J. D., Panageas, I., Piliouras, G., Simchowitz, M., Jordan, M. I., and Recht, B. First-order methods almost always avoid strict saddle points. *Mathematical Programming*, 176:311–337, 2019.

Lohe, M. A. Non-Abelian Kuramoto models and synchronization. *Journal of Physics A: Mathematical and Theoretical*, 42(39):395101, 2009.

Loshchilov, I., Hsieh, C.-P., Sun, S., and Ginsburg, B. nGPT: Normalized transformer with representation learning on the hypersphere. In *International Conference on Learning Representations (ICLR)*, 2025.

Mann, W. R. Mean value methods in iteration. *Proceedings of the American Mathematical Society*, 4(3):506–510, 1953.

Mardia, K. V. and Jupp, P. E. Directional statistics. *Wiley Series in Probability and Statistics*, 2000.

Miyato, T., Löwe, S., Geiger, A., and Welling, M. Artificial Kuramoto oscillatory neurons. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.13821.

Movahedi, S., Milovanović, V., Feigin, S. L., Theus, A., Hofmann, T., Boeva, V., Rusch, T. K., and Orvieto, A. Fixed-point reasoners: Stable and adaptive deep looped transformers, 2026. arXiv:2606.18206.

Nocedal, J. and Wright, S. J. *Numerical Optimization*. Springer, New York, 2nd edition, 2006.

Ramsauer, H., Schäfl, B., Lehner, J., Seidl, P., Widrich, M., Adler, T., Gruber, L., Holzleitner, M., Pavlović, M., Sandve, G. K., Greiff, V., Kreil, D., Kopp, M., Klambauer, G., Brandstetter, J., and Hochreiter, S. Hopfield networks is all you need. In *International Conference on Learning Representations (ICLR)*, 2021.

Rusch, T. K. and Mishra, S. Coupled oscillatory recurrent neural network (coRNN): An accurate and (gradient) stable architecture for learning long time dependencies. In *International Conference on Learning Representations (ICLR)*, 2021.

Scellier, B. and Bengio, Y. Equilibrium propagation: Bridging the gap between energy-based models and backpropagation. *Frontiers in Computational Neuroscience*, 11:24, 2017.

Strogatz, S. H. From Kuramoto to Crawford: Exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1–4): 1–20, 2000.

Trotter, H. F. On the product of semi-groups of operators. *Proceedings of the American Mathematical Society*, 10 (4):545–551, 1959.

Wang, G. et al. Hierarchical reasoning model, 2025. arXiv:2506.21734.

Wang, T. and Roychowdhury, J. OIM: Oscillator-based Ising machines for solving combinatorial optimisation problems. *Unconventional Computation and Natural Computation (UCNC), LNCS 11493*, pp. 232–256, 2019.

Xiao, M., Wang, Y., Han, D., Shan, C., and Li, D. Kuramoto oscillatory phase encoding: Neuro-inspired synchronization for improved learning efficiency, 2026. arXiv:2604.07904.

A. Mathematical Preliminaries

This appendix collects the mathematics needed to read both models. Standard references are Absil et al. (2008) for optimization on manifolds, Kuramoto (1984); Strogatz (2000) for coupled oscillators, and Lohe (2009) for Kuramoto models on higher-dimensional spheres.

A.1. Spheres, Tangent Spaces and Products of Spheres

Differentiating $\langle z(t),\, z(t)\rangle = 1$ along a curve on $\mathbb{S}^{n-1}$ gives $\langle \dot z,\, z\rangle = 0$, so the tangent space at z is the hyperplane $\mathrm{T}_z\mathbb{S}^{n-1} = \{v : \langle v,\, z\rangle = 0\}$. The projector $\mathrm{P}_z = \mathrm{I} - zz^\top$ of (1) is symmetric and idempotent, satisfies $\mathrm{P}_z\, z = 0$, and leaves tangent vectors unchanged. The radial part of a vector changes only its length and is invisible after normalization; only the tangential part moves a point along the sphere. This decomposition underlies every identity in the paper.

A state of M unit vectors $z = (z_1, \dots, z_M)$, $z_m \in \mathbb{S}^{N-1}$, lives on the product manifold $(\mathbb{S}^{N-1})^M$ of dimension $M(N-1)$, whose tangent space is the product of the factors' tangent spaces; the projector acts block by block. AKOrN uses $M = C/N$ small factors per position; SFPRM uses one factor per token ($N = d$), so its state space is $\mathcal{S} = (\mathbb{S}^{d-1})^T$. Rescaling by $1/\sqrt{M}$ embeds $(\mathbb{S}^{N-1})^M$ into $\mathbb{S}^{MN-1}$ as the subset on which all blocks have equal norm. On a sphere a state can neither grow nor vanish, so iterating a map on it is bounded by construction and only directions carry information; FPRM enforces boundedness by coupling residual coefficients (Movahedi et al., 2026), whereas SFPRM obtains it from the geometry (Anonymous, 2026).

The geodesic distance is the angle $\theta = \angle(u, v)$. SFPRM's halting test uses the chordal quantity $1 - \langle u,\, v\rangle = 1 - \cos\theta = \frac{1}{2}\|u - v\|^2$. Geodesics are great circles; the one leaving u in unit tangent direction e is $t \mapsto \cos(t)u + \sin(t)e$, which gives Exp_u. Its inverse Log_u follows from $\|\mathrm{P}_u\, v\|^2 = 1 - \cos^2\theta = \sin^2\theta$. Slerp is undefined at antipodal pairs, where the shortest arc is not unique.

A.2. Retractions and Riemannian Gradients

A retraction is a smooth map $R_z : \mathrm{T}_z\mathbb{S}^{n-1} \to \mathbb{S}^{n-1}$ with $R_z(0) = z$ and identity derivative at 0, so it agrees with Exp_z to first order. Normalization, $R_z(v) = \Pi(z + v)$, is the retraction used by AKOrN and nGPT. For tangent v, $\|z + v\|^2 = 1 + \|v\|^2$, hence

$$\Pi(z + v) = \cos\varphi\, z + \sin\varphi\, \tfrac{v}{\|v\|}, \qquad \tan\varphi = \|v\| : \tag{36}$$

normalization moves along the same great circle as $\mathrm{Exp}_z(v)$ but by the angle $\arctan\|v\| < \pi/2$ instead of $\|v\|$. Retraction, exponential map and Slerp therefore differ only in how a step length is converted into an angle, which is why step sizes can be traded between them exactly.

For a smooth f on $\mathbb{R}^n$, the Riemannian gradient of its restriction to the sphere is $\mathrm{grad}\, f(z) = \mathrm{P}_z\, \nabla f(z)$, and Riemannian gradient descent is $z \leftarrow R_z(-\gamma\, \mathrm{grad}\, f(z))$. For the linear potential $f(z) = -\langle c,\, z\rangle$, $\mathrm{grad}\, f = -\mathrm{P}_z\, c$ (AKOrN's stimulus); for the geodesic potential $f(z) = \frac{1}{2}\angle(z, x)^2$, $\mathrm{grad}\, f = -\mathrm{Log}_z\, x$ (SFPRM's injection, Proposition 5.5).

A.3. Antisymmetric Matrices and Rotations

If $\Omega^\top = -\Omega$ then $\langle \Omega z,\, z\rangle = 0$, so Ωz is tangent and the flow $\dot z = \Omega z$ preserves the norm; its solution $e^{t\Omega} z(0)$ is a rotation. In a suitable orthonormal basis Ω is block diagonal with 2×2 blocks $\omega_k R$, $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $R^2 = -\mathrm{I}$, plus a zero row and column when N is odd; each block rotates one plane at the natural frequency ω_k. For odd N a rotation fixes an axis, and by the hairy-ball theorem every tangent field on $\mathbb{S}^{N-1}$ vanishes somewhere. For even N, Ω can have no zero eigenvalue and $\dot z = \Omega z$ moves every point. This is why AKOrN uses even N, and it is the obstruction to fixed-point existence that SFPRM's injection repairs.

A.4. The Kuramoto Model in Vector Form

In $\dot\theta_i = \omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i)$ (Kuramoto, 1984; Strogatz, 2000), positive couplings synchronize phases and negative couplings repel them; the order parameter $re^{i\psi} = \frac{1}{n}\sum_j e^{i\theta_j}$ measures synchrony. A phase-locked state rotates at a common frequency with constant phase differences and is a fixed point in the co-rotating frame. With $x_i = (\cos\theta_i, \sin\theta_i)$ we have $\dot x_i = \dot\theta_i R x_i$, where $Rx_i = (-\sin\theta_i, \cos\theta_i)$ is the unit tangent at x_i, and $\langle x_j,\, Rx_i\rangle = \sin(\theta_j - \theta_i)$. Hence $\mathrm{P}_{x_i}\, x_j = \sin(\theta_j - \theta_i) R x_i$, which gives (5). Replacing $\omega_i R$ by an antisymmetric Ω_i, scalar couplings by matrices J_{ij}, and

adding an external field c_i that pins oscillators as an injected signal locks a physical oscillator (Adler, 1946) yields (6) (Lohe, 2009; Miyato et al., 2025). The AKOrN implementation rotates with the opposite sign, which only flips the sign convention of ω.

A.5. Gradient Flows and Lyapunov Functions

If $c + Jx = -\nabla E(x)$ and $\Omega = 0$, then (6) is the Riemannian gradient flow $\dot{x} = -\operatorname{grad} E(x)$, along which

$$\tfrac{d}{dt} E(x(t)) = \langle \nabla E,\, \dot{x} \rangle = -\|\operatorname{grad} E(x)\|^2 \le 0, \tag{37}$$

because $\dot{x}$ is tangent. E is a Lyapunov function, and on a compact state space the flow settles into the set of stationary points (LaSalle's invariance principle). For $E(x) = -\frac{1}{2} x^\top J x - c^\top x$, the condition $c + Jx = -\nabla E$ holds exactly when J is symmetric, which is the origin of (8). A rotation Ωx is not a gradient, and neither is a non-symmetric coupling such as softmax attention; in those cases no energy is guaranteed.

A.6. Fixed-Point Iterations and Damping

A looped model iterates $z_{i+1} = F(z_i)$ and reads its answer at a fixed point $z^\star = F(z^\star)$. If F is a contraction with constant $q < 1$, Banach's theorem gives a unique fixed point and $\|z_i - z^\star\| \le q^i \|z_0 - z^\star\|$; this is the form of FPRM's global result (Movahedi et al., 2026). Near a fixed point the error evolves as $e_{i+1} \approx J e_i$ with $J = \mathrm{d}F(z^\star)$. The damped (Krasnosel'skii–Mann) iteration $z_{i+1} = (1-\eta) z_i + \eta F(z_i)$ (Mann, 1953) has the same fixed points for every $\eta \in (0, 1]$ and linearizes to $(1-\eta)\mathrm{I} + \eta J$, mapping each eigenvalue ζ to $1 + \eta(\zeta - 1)$. Since $|1 + \eta(\zeta-1)|^2 = 1 - 2\eta(1 - \operatorname{Re}\zeta) + \eta^2 |\zeta - 1|^2$, the mode decays exactly when $\operatorname{Re}\zeta < 1$ and $\eta < 2(1 - \operatorname{Re}\zeta)/|\zeta - 1|^2$. Damping tames rotating or overshooting modes but no step size helps a mode with $\operatorname{Re}\zeta \ge 1$. On the sphere, $\operatorname{Slerp}_\eta(z_i, F(z_i))$ replaces linear interpolation with the same fixed points and linearization (Anonymous, 2026).

By the implicit function theorem, the parameter gradient at $z^\star$ involves $(\mathrm{I} - J)^{-1}$. FPRM and SFPRM avoid forming it: they backpropagate through windows of K iterations, detach the state, and apply a loss after every window (deep supervision with truncated backpropagation through time). Under contraction this corresponds to truncating the Neumann series $\sum_{k<K} J^k$ for $(\mathrm{I} - J)^{-1}$ (Movahedi et al., 2026).

A.7. Transformers on the Sphere

nGPT (Loshchilov et al., 2025) folds a sub-layer output $G(h)$ into the state by $h \leftarrow \operatorname{Norm}(h + \alpha \odot (\operatorname{Norm}(G(h)) - h))$ with per-coordinate eigen learning rates α; for scalar α this is normalized linear interpolation toward $\operatorname{Norm}(G(h))$. Softmax attention,

$$\operatorname{Attn}(h)_t = \sum_s a_{ts}(h)\, W_O W_V h_s, \qquad a_{ts}(h) = \operatorname{softmax}_s \big(s_{qk} \langle \operatorname{Norm}(W_Q h_t),\, \operatorname{Norm}(W_K h_s) \rangle \big), \tag{38}$$

is a linear map of the token vectors with state-dependent weights, i.e. a coupling $J_{ts}(h) = a_{ts}(h) W_O W_V$ of the kind AKOrN uses as attention connectivity.

A.8. Operator Splitting

For smooth vector fields A, B with flows $\phi^A_\varepsilon, \phi^B_\varepsilon$, $\phi^B_\varepsilon \circ \phi^A_\varepsilon = \phi^{A+B}_\varepsilon + O(\varepsilon^2)$, with leading error proportional to the Lie bracket $[A, B]$ (Trotter, 1959). The same holds for single retraction steps $z \mapsto R_z(\varepsilon A(z))$, which agree with their flows to first order. A model that applies stimulus, coupling and nonlinearity as separate small steps is therefore, to first order, a single step with the summed drive (Theorem 5.8).

B. Proofs and Additional Statements

Attention sub-layer with per-coordinate eigen learning rates. The following restates Proposition 5.6 and adds the per-coordinate case.

Proposition B.1 (Attention sub-layer, full statement)**.** *Let* $A_t := \operatorname{Attn}(h)_t \neq 0$, $\hat{A}_t := A_t / \|A_t\|$ *and* $J_{ts}(h)$ *as in* (13).

(a) If $\alpha_A = a\mathbf{1}$ *and* $1 + a(\langle \hat{A}_t,\, h_t \rangle - 1) > 0$ *(automatic for* $a < 1/2$*), then* (11) *equals* $h_t \leftarrow \Pi\big(h_t + \gamma^A_t \operatorname{P}_{h_t} \sum_s J_{ts}(h) h_s\big)$ *with* $\gamma^A_t = a / \big(\|A_t\|(1 + a(\langle \hat{A}_t,\, h_t \rangle - 1))\big)$.

(b) If $\alpha_A \in \mathbb{R}^d$, $w_t := \alpha_A \odot (\hat{A}_t - h_t)$ *and* $1 + \langle w_t,\, h_t\rangle > 0$, *then* (11) *equals*

$$h_t \leftarrow \Pi\Big(h_t + \tfrac{1}{1+\langle w_t,\, h_t\rangle}\, \mathrm{P}_{h_t} \Big[\tfrac{1}{\|A_t\|}\, \mathrm{diag}(\alpha_A) \sum_s J_{ts}(h) h_s - \mathrm{diag}(\alpha_A) h_t\Big]\Big) : \tag{39}$$

the coupling is diagonally preconditioned and a self-drive $-\mathrm{diag}(\alpha_A)h_t$ *appears, whose tangent part is* $-\operatorname{grad} \frac{1}{2} h^\top \mathrm{diag}(\alpha_A) h$ *and vanishes when* α_A *is constant.*

Proof. Apply Lemma 4.1 with $z = h_t$ and $w = w_t$. For (a), $\mathrm{P}_{h_t} w_t = a\, \mathrm{P}_{h_t} \hat{A}_t = \frac{a}{\|A_t\|}\, \mathrm{P}_{h_t} \sum_s J_{ts} h_s$ because $\mathrm{P}_{h_t} h_t = 0$, and $1 + \langle w_t,\, h_t\rangle = 1 + a(\langle \hat{A}_t,\, h_t\rangle - 1)$. For (b), P_{h_t} no longer annihilates $\mathrm{diag}(\alpha_A)h_t$, which produces the self-drive. □

Proposition B.2 (nGPT MLP sub-layer)**.** *With* $M_t := \mathrm{MLP}(h)_t \neq 0$ *and* $\hat{M}_t := M_t / \|M_t\|$, (12) *is the step of Proposition B.1 with* $\sum_s J_{ts} h_s$ *replaced by the token-wise nonlinear drive* M_t.

Proof. Identical to Proposition B.1 with A_t replaced by M_t. □

Proof of Theorem 5.8 (sketch). For a smooth tangent field v, $\Pi(y + \varepsilon v(y)) = y + \varepsilon v(y) + O(\varepsilon^2)$. By Lemma 4.2, $\mathrm{Slerp}_{\varepsilon\kappa_0}(z, x) = \mathrm{Exp}_z(\varepsilon\kappa_0 \operatorname{Log}_z x) = z + \varepsilon\kappa_0 \frac{\theta^x}{\sin\theta^x}\, \mathrm{P}_z\, x + O(\varepsilon^2)$. Each nGPT sub-layer gives $h \mapsto h + \varepsilon\kappa\, \mathrm{P}_h\, \hat{A} + O(\varepsilon^2)$ (or $\hat{M}$). Since every intermediate state equals $z + O(\varepsilon)$, evaluating all drives at z costs $O(\varepsilon^2)$ (Lie–Trotter). Hence $\tilde{z} = z + \varepsilon\, \mathrm{P}_z\, V(z) + O(\varepsilon^2)$ and $\theta_i = O(\varepsilon)$. Then Lemma 4.2 gives $\gamma_\eta(\theta_i) = \eta + O(\varepsilon^2)$ and $z_{i+1} = \Pi(z + \eta\, \mathrm{P}_z\, \tilde{z}) + O(\varepsilon^2) = \Pi(z + \eta\varepsilon\, \mathrm{P}_z\, V(z)) + O(\varepsilon^2)$. □

Proof of Proposition 5.10. Write $v(z) := \Omega z + \mathrm{P}_z\, D(z)$, so $v(z^\star) = 0$. Terms involving the derivative of γ multiply $v(z^\star)$ and vanish. For a tangent perturbation u, $\mathrm{d}(\mathrm{P}_z\, D)[u] = \mathrm{P}_z\, J_D u - \langle D,\, u\rangle z - \langle D,\, z\rangle u$. The middle term is normal and is removed by the derivative of Π at $z^\star$, which equals P because $z^\star + \gamma v(z^\star) = z^\star$ has unit norm. Collecting terms gives the claim. □

Proof of Corollary 5.11. At a fixed point of SFPRM, $F_\theta(z^\star; x) = z^\star$, so $\mu_t = \langle z_t^\star,\, z_t^\star\rangle = 1$ and $\mathrm{P}\, J_F\, \mathrm{P} - \mathrm{I}$ restricted to the tangent space equals $J - \mathrm{I}$. With $\Omega = 0$ and $\gamma \to \eta$, Proposition 5.10 gives $\mathrm{I} + \eta(J - \mathrm{I}) = (1 - \eta)\mathrm{I} + \eta J$, the linearization of the Slerp step reported by Anonymous (2026). □

Proof of Proposition 7.3. Fixed points satisfy $\Omega x + h - \mu x = 0$ with $\mu := \langle h,\, x\rangle$. In plane k, $\Omega_k = \omega_k R'$ with $R' = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, $R'^2 = -\mathrm{I}$, and $(\omega_k R' - \mu \mathrm{I})^{-1} = -(\omega_k R' + \mu \mathrm{I})/(\omega_k^2 + \mu^2)$. Hence $x_k = (\omega_k R' + \mu \mathrm{I}) h_k / (\omega_k^2 + \mu^2)$, and since $R' h_k \perp h_k$ and $\|R' h_k\| = \|h_k\|$,

$$\|x\|^2 = g(\mu) := \sum_k \frac{\|h_k\|^2}{\omega_k^2 + \mu^2}. \tag{40}$$

Consistency holds automatically: $\langle h,\, x\rangle = \sum_k \mu \|h_k\|^2 / (\omega_k^2 + \mu^2) = \mu\, g(\mu)$, which equals μ whenever $g(\mu) = 1$. Fixed points therefore correspond to solutions of $g(\mu) = 1$. The function g is even, strictly decreasing in $|\mu|$, tends to 0 as $|\mu| \to \infty$, and satisfies $g(0) = \sum_k \|h_k\|^2 / \omega_k^2$, so a solution exists if and only if $g(0) \geq 1$. □

C. Numerical Verification Setup

All checks used NumPy in double precision.

- **Lemmas 4.1 and 4.2.** 5,000 random draws in $d = 64$: unit z and v, damping $\eta \sim \mathcal{U}(0.02, 1)$, and nGPT-style updates $w = \alpha \odot (A - z)$ with unit A and $\alpha_i \sim \mathcal{U}(0, 0.5)$; only draws satisfying the positivity conditions enter the maxima.
- **Proposition 5.6.** $T = 6$ tokens, $d = 16$, $H = 2$ heads with QK-normalization (logit scale 4), scalar eigen learning rate $a = 0.3$. The identity $\mathrm{Attn}(h)_t = \sum_s J_{ts}(h) h_s$ holds to 2.2×10^{-16}.
- **Theorem 5.8.** The same tokens with one nGPT-style layer (attention and an MLP $h \mapsto \tanh(W_1 h)$), $\eta = 1$, all $\kappa = 1$; the error between the split iteration and the single step is reported at $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$.

- **Proposition 5.10.** One token in $d = 6$ with $F(z) = \Pi(\tanh(Wz + b) + 1.2z)$, $\gamma = 0.5$, and either $\Omega = 0$ or three rotation planes with frequencies $(0.3, 0.2, 0.1)$. Fixed points were found by iterating 50,000 steps (residual $< 10^{-16}$); Jacobians were compared with central differences of step 10^{-6} on the tangent space.

- **Proposition 7.3.** $N = 4$ with plane frequencies $(1.0, 0.6)$, random drives rescaled so that $\sum_k \|h_k\|^2/\omega_k^2 \in \{0.6, 0.9, 0.99, 1.01, 1.1, 2\}$, step $\gamma = 0.05$, 40,000 steps from three random initializations. For values ≥ 1.01 the final tangent speed was below 10^{-15}; for values ≤ 0.99 it stayed above 7×10^{-2}.

D. Proofs for the Energy-Based SFPRM

Proof of Proposition 8.2. Write $q_t^k = W_Q^k u_t$, $k_s^k = W_K^k z_s$ and $\ell_{ts}^k = \beta_k \langle q_t^k, k_s^k \rangle$. For the Euclidean gradient of E_A with respect to u_t only the terms of head k in which t is the query index contribute, and $\partial_{u_t}\big[-\frac{\sigma_k}{\beta_k}\log\sum_{s\neq t} e^{\ell_{ts}^k}\big] = -\sigma_k \sum_{s\neq t} a_{ts}^k (W_Q^k)^\top k_s^k = -\sigma_k \sum_{s\neq t} a_{ts}^k M_k z_s$. For z_s only terms in which s is the key index contribute, one for every query $t \neq s$, giving $-\sigma_k \sum_{t\neq s} a_{ts}^k (W_K^k)^\top q_t^k = -\sigma_k \sum_{t\neq s} a_{ts}^k M_k^\top u_t$; renaming indices yields the key-path sum in (31). The bilinear term gives $-\kappa_C C^\top u_t$ and $-\kappa_C C z_t$, the memory term gives $-\kappa_M \sum_\mu \mathrm{softmax}_\mu(\beta_M \Xi u_t)\,\xi_\mu$, and the injection term gives $-\kappa_0 \operatorname{Log}_{z_t} x_t$ after projection (Proposition 5.5). The Riemannian gradient on a product of spheres is the token-wise projection P_{w_t} of the Euclidean gradient (Section A), which proves (31)–(32). In the flat variant $u_t = z_t$ and the two attention contributions to token t add to $\sum_{s\neq t}\big(a_{ts}^k M_k + a_{st}^k M_k^\top\big) z_s$, so $J_{ts} = \sum_k \sigma_k (a_{ts}^k M_k + a_{st}^k M_k^\top)$ and $J_{st}^\top = \sum_k \sigma_k (a_{st}^k M_k^\top + a_{ts}^k M_k) = J_{ts}$. □

Proof of Theorem 8.3. *Existence of L.* E_A, E_C and E_M are real-analytic on $\mathbb{R}^{2Td}$, so their restrictions to the compact manifold $(\mathbb{S}^{d-1})^{2T}$ have bounded Riemannian Hessians, and for the exponential retraction $t \mapsto E(\mathrm{Exp}_w(tv))$ has second derivative $\mathrm{Hess}\, E(\gamma(t))[\dot\gamma, \dot\gamma] \leq \lambda_{\max}(\mathrm{Hess}\, E)\|v\|^2$, which gives the inequality with $L = \sup \lambda_{\max}(\mathrm{Hess}\, E)$ by Taylor's theorem (Boumal, 2023, Ch. 10). For the injection term, on $\mathbb{S}^{d-1}$ the Hessian of $\frac{1}{2}\angle(\cdot, x)^2$ at angle $\theta \in (0, \pi)$ has eigenvalue 1 along the geodesic to x and $\theta\cot\theta \leq 1$ on its orthogonal complement; at the antipode the function has a concave corner, so its second derivative along any geodesic is bounded above by 1 in the distributional sense and the inequality holds with the contribution κ_0 at every w with no token antipodal to its input, which is the exceptional set of the theorem; for the chordal injection $\sum_t (1 - \langle z_t, x_t\rangle)$, AKOrN's stimulus potential, the Hessian is bounded by 1 everywhere and no exception is needed. *(i)* Take $v = -\eta\,\mathrm{grad}\, E(w_i)$: $E(w_{i+1}) \leq E(w_i) - \eta\|\mathrm{grad}\, E(w_i)\|^2 + \frac{L}{2}\eta^2\|\mathrm{grad}\, E(w_i)\|^2 = E(w_i) - c_\eta\|\mathrm{grad}\, E(w_i)\|^2$. *(ii)* Summing (i) over $i < N$ and using that E is continuous on a compact set, hence bounded below by $\min E$, gives $c_\eta \sum_{i<N}\|\mathrm{grad}\, E(w_i)\|^2 \leq E(w_0) - \min E$ for every N. Hence $\mathrm{grad}\, E(w_i) \to 0$, and by continuity of $\mathrm{grad}\, E$ every accumulation point is stationary. If the halting test has not fired before iteration N then $\|\mathrm{grad}\, E(w_i)\| \geq \max_t \|\mathrm{grad}_{w_t} E(w_i)\| \geq \tau_g$ for all $i < N$, so $N c_\eta \tau_g^2 \leq E(w_0) - \min E$. *(iii)* By (i) and $v = -\eta\,\mathrm{grad}\, E$, the iteration is a descent method with exact gradient direction and sufficient decrease, so the Łojasiewicz argument of Absil et al. (2005), applied in a chart around an accumulation point where E is real-analytic, shows that the sequence has finite length and converges. For decaying step sizes $\eta_i \in [\eta_{\min}, \eta]$, (i) holds with $c_{\eta_i} \geq \min(c_{\eta_{\min}}, c_\eta)$ because c is concave in η on $(0, 2/L)$. □

Proof of Proposition 8.4. F_E is the generalized AKOrN step of Definition 5.1 with $\Omega = 0$, $\gamma = 1$, drive $D = -\nabla E$ and the exponential retraction in place of normalization; at a fixed point the two retractions have the same derivative because both equal the identity on the tangent space to first order and the step vanishes. Proposition 5.10 with $J_D = -\nabla^2 E$ and $\mu_t = \langle D_t, w_t^\star\rangle = -\langle \nabla_{w_t} E, w_t^\star\rangle$ gives $\mathrm{d}F_E = \mathrm{I} - \big(\mathrm{P}\,\nabla^2 E\,\mathrm{P} - \mathrm{M}_{\mu'}\big)$ with $\mu_t' = \langle \nabla_{w_t} E, w_t^\star\rangle$, and the bracket is the Riemannian Hessian of E on the product of spheres (Absil et al., 2008, Sec. 5.4). It is self-adjoint on the tangent space because $\mathrm{P}\,\nabla^2 E\,\mathrm{P}$ is symmetric and $\mathrm{M}_{\mu'}$ is a token-wise multiple of the identity. By Corollary 5.11 the Slerp step linearizes to $(1-\eta)\mathrm{I} + \eta\,\mathrm{d}F_E = \mathrm{I} - \eta\,\mathrm{Hess}\, E$. The spectral theorem gives real eigenvalues h of the Hessian and $\zeta = 1 - h$ of $\mathrm{d}F_E$. The mode decays iff $|1 - \eta h| < 1$ iff $0 < h < 2/\eta$; all modes decay iff $\mathrm{Hess}\, E \succ 0$ and $\eta < 2/\lambda_{\max}$, and for real ζ the window $2(1 - \mathrm{Re}\,\zeta)/|\zeta - 1|^2$ of Corollary 5.11 equals $2/h$. If some $h < 0$ then $|1 - \eta h| > 1$ for every $\eta > 0$. □

Eigen learning rates as a preconditioner. Let $D = \mathrm{diag}(\alpha) \succ 0$ be fixed and replace the step by $w \leftarrow \mathrm{Exp}_w(-\eta\ \mathrm{P}\ D\ \mathrm{grad}\, E(w))$. Then $\langle \mathrm{grad}\, E, \mathrm{P}\, D\, \mathrm{grad}\, E\rangle = \langle \mathrm{grad}\, E, D\,\mathrm{grad}\, E\rangle \geq \alpha_{\min}\|\mathrm{grad}\, E\|^2$, so the direction is a descent direction and Theorem 8.3 holds with c_η replaced by $\eta(\alpha_{\min} - \eta L \alpha_{\max}^2/2)$. At a stationary point the Jacobian is $\mathrm{I} - \eta(\mathrm{P}\, D\, \mathrm{P})\,\mathrm{Hess}\, E$, and $(\mathrm{P}\, D\, \mathrm{P})\,\mathrm{Hess}\, E$ restricted to the tangent space is similar to $S^{1/2}\,\mathrm{Hess}\, E\, S^{1/2}$ with $S = \mathrm{P}\, D\, \mathrm{P} \succ 0$ on the tangent space, which is symmetric, so the spectrum stays real.

Proof of Proposition 8.5. The residual bound is Definition 8.1. Energy selection with $\sigma_k = \kappa_C = \kappa_M = 0$ is anchor-angle selection. Let $w^\sharp$ be the global minimizer with basin $\mathcal{B}$ of measure $p > 0$. A restart in $\mathcal{B}$ halts at a τ_g-stationary point $w^{(b)}$ in the basin, with $E(w^{(b)}) \le \min E + \delta(\tau_g)$, where the halting slack δ vanishes with τ_g. Any other halted state is within the same slack of a stationary value, which by assumption exceeds $\min E + 2\delta$ (the gap condition of the proposition), so $w^{(b)}$ has the smallest energy among the halted states and decodes correctly. The probability that none of B starts falls in $\mathcal{B}$ is $(1-p)^B \to 0$. □

Implicit gradient and equilibrium propagation. At an attracting halted state, $\operatorname{grad} E(w^\star;\theta) = 0$ with $\operatorname{Hess} E(w^\star) \succ 0$, so by the implicit function theorem (in a chart) $w^\star(\theta)$ is smooth and differentiating the stationarity condition gives $\operatorname{Hess} E \frac{\mathrm{d}w^\star}{\mathrm{d}\theta} + \frac{\partial \operatorname{grad} E}{\partial \theta} = 0$. Hence $\frac{\mathrm{d}\ell}{\mathrm{d}\theta} = \langle \operatorname{grad} \ell, \frac{\mathrm{d}w^\star}{\mathrm{d}\theta}\rangle = -\langle \operatorname{Hess} E^{-1} \operatorname{grad} \ell, \partial_\theta \operatorname{grad} E\rangle$, which is (34). With $J = \mathrm{d}F_E = \mathrm{I} - \operatorname{Hess} E$ the matrix $(\mathrm{I}-J)^{-1}$ of Section A is $\operatorname{Hess} E^{-1}$, and the truncated Neumann series of truncated BPTT becomes $\eta \sum_{k<K} (\mathrm{I} - \eta \operatorname{Hess} E)^k$, whose error after K terms is $(1-\eta\lambda_{\min})^K$, i.e. $K = O(\kappa \log(1/\epsilon))$ for $\eta = 1/\lambda_{\max}$, against $O(\sqrt{\kappa}\log(1/\epsilon))$ for conjugate gradient (Nocedal & Wright, 2006, Thm. 5.5). For (35), let $G(\theta,\beta) := \min_w E(w;\theta) + \beta\ell(w) = E(w^\star_\beta;\theta) + \beta\ell(w^\star_\beta)$. By the envelope theorem $\partial_\theta G = \partial_\theta E(w^\star_\beta;\theta)$ and $\partial_\beta G = \ell(w^\star_\beta)$; equality of mixed partial derivatives gives $\frac{\mathrm{d}}{\mathrm{d}\theta}\ell(w^\star_\beta) = \frac{\mathrm{d}}{\mathrm{d}\beta}\partial_\theta E(w^\star_\beta;\theta)$, and $\beta = 0$ yields (35) (Scellier & Bengio, 2017); the argument uses only stationarity on the manifold.

E. Pilot Experiment: Energy-Based versus Standard SFPRM

Tasks. Sudoku of size 4×4 (2×2 boxes, $T = 16$) and 6×6 (2×3 boxes, $T = 36$). Puzzles are generated by filling a random grid with a randomized backtracking solver and removing cells in random order while the puzzle keeps a unique solution, until a target number of clues is reached. For 6×6, training uses 12,000 puzzles with 14–22 clues (mean 18.0); the in-distribution (ID) test set has puzzles from the same range and the out-of-distribution (OOD) test set puzzles with 8–13 clues (mean 11.1), mirroring the easy/hard split used for AKOrN. For 4×4, training uses 400 puzzles with 7–10 clues and the OOD test set has 4–6 clues; since only 288 distinct 4×4 grids exist, test grids can coincide with training grids, and the 4×4 task measures constraint propagation rather than generalization across grids. Evaluation uses 500 puzzles per test set. Each cell is a token; the input embedding is $x_t = \mathrm{Norm}(e_{\mathrm{digit}(t)} + W p_t)$ with a learned digit table (blank included) and a learned linear map W of the row/column/box one-hot features p_t, shared by both models. The readout is linear, $s\, W_{\mathrm{out}} z_t + b$, and the loss is the cross-entropy over all cells.

Models. *SFPRM*: $d = 64$, $D = 2$ nGPT layers with $H = 4$ heads, QK-normalized attention with a learned logit scale per head, a GELU MLP of width $4d$, per-coordinate eigen learning rates initialized at 0.25, injection $\lambda = 0.5$, Slerp damping; PSF parameters. *Energy SFPRM* (Definition 8.1, hidden variant): $d = 64$, $H = 8$ heads of dimension 8 with signed weights initialized at ±0.2 (alternating), W_Q, W_K with entries of standard deviation $1/\sqrt{d}$, logit scales $\beta_k = d/\sqrt{d_h}$, $K = 1024$ memories with $\beta_M = \sqrt{d}$, $\kappa_0 = 0.5$, $\kappa_C = 0.5$, $\kappa_M = 0.1$, geodesic injection; PEN parameters. All positive scales are softplus-parametrized and learned.

Training. Both models are trained identically: AdamW (learning rate 2×10^{-3}, weight decay 0.01 on matrices, cosine decay, 200 warm-up steps, gradient clipping at 1), batch 128, 4,000 steps on 6×6 (400 for SFPRM and 2,000 for the energy model on 4×4, each run until its training accuracy had saturated), random unit-norm initial states, deep supervision with two windows of $K = 8$ iterations at $\eta = 1$ and the state detached between windows, exactly the recipe described for FPRM and SFPRM. For the energy model each training iteration uses Armijo backtracking on E (acceptance constant 10^{-4}, at most four halvings, step size treated as a constant), which keeps the training dynamics inside the descent regime of Theorem 8.3; without it, energy increases in up to half of the iterations once the attention sharpens, and the model does not learn.

Evaluation. At test time both models run the damped loop from random unit-norm states for at most 128 iterations with the chordal halting test at $\tau_c = 10^{-4}$, η starting at 1 and halved after 3 iterations without residual improvement (patience rule, minimum $1/64$); for the energy model we report Armijo backtracking in place of the patience rule, and use it for the spectrum and restart measurements as well; the patience rule, which starts at $\eta = 1$, far outside the energy model's window, halts only about half of the puzzles within 128 iterations. *Iterations to halt* are counted over puzzles that halt. *Spectrum*: for 20 ID and 20 OOD puzzles the halted state is polished in double precision (continuing the damped iteration until the residual is below 10^{-12} or for 2,000 steps), and the Jacobian of the proposal map F on the tangent space is computed by automatic differentiation in an orthonormal tangent basis; we report the fraction of eigenvalues with $|\operatorname{Im}\zeta| > 10^{-8}$, $\max|\operatorname{Im}\zeta|$, the

relative asymmetry $\|J - J^\top\|_{\max}/\|J\|_{\max}$, and the damping window of Corollary 5.11. *Restarts*: for $B \in \{1, 2, 4, \dots, 64\}$ independent random starts per puzzle we select one halted state by (a) the final energy $E(w^\star; x)$, (b) the energy summed over the first 32 iterations, as AKOrN does, (c) the anchor-angle potential $\Phi_x(z^\star)$, which is available in plain SFPRM, (d) majority vote over decoded boards, and report board accuracy; the oracle is the fraction of puzzles for which any of the B restarts is correct.

Results of the first energy model. Table 5 and Figure 2 summarize the comparison of Section 8.5. Four observations. *Accuracy and training cost.* The energy model learns both tasks, but needs several times more optimization steps than SFPRM to leave the "clues only" plateau (Figure 3), and at an equal budget it is behind: on 4×4 it reaches 82.4% ID and 17.2% OOD against 97.8% and 65.6% for SFPRM, and on 6×6 it does not leave the plateau within the 4,000-step budget in which SFPRM reaches 99.8% ID and 89.2% OOD (cell accuracy 72%, no board solved). Without the three design choices of Section 8.2 and Armijo backtracking in training it does not leave the plateau at all. *Spectrum.* At every polished halted state the energy model's Jacobian is symmetric to machine precision and all eigenvalues are real, whereas 96.4% of SFPRM's eigenvalues are complex with imaginary parts up to 0.61 (Figure 2, left). The learned energy is stiff, however: $\lambda_{\max}(\mathrm{Hess}\, E) \approx 12.2$, so the admissible step is $\eta < 2/\lambda_{\max} \approx 0.16$ rather than SFPRM's ≈ 1.40, and the slow modes of a stiff Hessian make halting *slower*, not faster: 68 / 114 iterations (median / p90, OOD) against 19 / 45 for SFPRM. Real eigenvalues remove the rotational obstruction of Corollary 5.11, but the condition number of the learned Hessian decides the halting time, and nothing in the training objective controls it. *Halting rule.* With Armijo backtracking the energy model halts 81.8% of the OOD puzzles within 128 iterations and the energy never increases; the patience rule, which starts at $\eta = 1$, halts about half as many (Table 6). *Energy selection.* Energy voting does not help here: selecting the lowest-energy halted state among $B = 64$ restarts gives 12.4% OOD on 4×4, below the single-restart accuracy, while majority voting gives 20.6% and the oracle 36.0%. The premise of Proposition 8.5 fails, as the lowest minima of the learned energy are not the correct boards; deep supervision never asks them to be. Anchor-angle selection fails in the same way for both models.

Table 5. Pilot comparison on 4×4 and 6×6 Sudoku (500 test puzzles each; board accuracy in %; one restart unless stated; iterations to halt at $\tau_c = 10^{-4}$ over halted puzzles, median / 90th percentile; spectrum of $\mathrm{d}F$ at polished halted OOD states). The energy model uses Armijo backtracking at test time, SFPRM the patience rule.

	4×4		6×6	
	SFPRM	Energy	SFPRM	Energy
Parameters ($\times 10^3$)	101	79	101	80
Steps to 90% train acc.	200	1250	400	> 4000
Board acc. ID	97.8	82.4	99.8	0.0
Board acc. OOD	65.6	17.2	89.2	0.0
Halted, OOD (≤ 128 it.)	73.4	81.8	90.6	30.4
Iterations to halt, OOD	19 / 45	68 / 114	19 / 37	102 / 122
Complex ζ	96.4	0.0	96.4	0.0
$\max \lvert \mathrm{Im}\, \zeta \rvert$	0.61	0	0.65	0.02
$\lambda_{\max}(\mathrm{Hess}\, E)$	–	12.2	–	17.9
Window (Corollary 5.11)	1.40	0.16	1.33	0.12
OOD, B=64: energy	–	12.4	–	0.0
OOD, B=64: anchor	48.8	20.4	92.8	0.0
OOD, B=64: majority	67.0	20.6	90.6	0.0
OOD, B=64: oracle	81.8	36.0	93.8	0.0

Full results. Table 6 lists the ID and OOD numbers for both tasks and both halting rules, and Figure 3 shows the training curves and the distribution of iterations to halt. On 4×4 the energy model solves 82.4% of ID and 17.2% of OOD boards after 2,000 steps (SFPRM: 97.8% and 65.6% after 400), halting 93.6% and 81.8% of the puzzles within 128 iterations under Armijo backtracking against 47% and 28% under the patience rule. On 6×6 its training cell accuracy rises from 58% to 73% over 4,000 steps but no board is solved, so the restart curves of Figure 4 are flat at zero; the spectrum and halting measurements at its halted states remain valid and show the same picture as on 4×4: real eigenvalues (the ID Jacobians have relative asymmetry $\leq 4 \times 10^{-2}$ and no complex pair; on OOD the polishing step did not reach the 10^{-12} residual within 2,000 iterations at the small admissible step for all puzzles, leaving imaginary parts up to 0.09 in one of six cases), a stiff Hessian with $\lambda_{\max} \approx 14$–$18$, a window of 0.12–0.16, and halting after 88 (ID) and 102 (OOD) iterations at the median, against 11 and 19 for SFPRM. The 4×4 restart curves show the selection rules in detail: energy selection

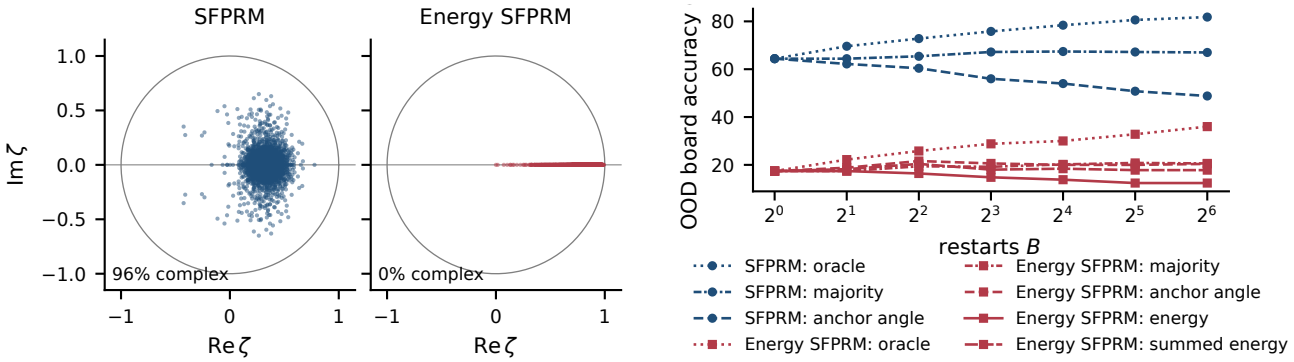


Figure 2. First energy model against SFPRM on 4×4 Sudoku. Left: eigenvalues of the tangent Jacobian at polished halted OOD states (three puzzles pooled); SFPRM at $\eta = 1$, the energy model for the damped map at $\eta = 1/\lambda_{\max}(\mathrm{Hess}\, E)$, half its window. Right: OOD board accuracy against the number of restarts B under the selection rules of Proposition 8.5.

decreases monotonically with B (84.8% at $B = 1$ to 78.8% at $B = 64$ ID; 17.4% to 12.4% OOD), the summed energy and the anchor angle are flat, majority voting gains little, and the oracle rises to 95.2% ID and 36.0% OOD, so the restarts do contain correct boards that no available score identifies.

Table 6. Full pilot results (board accuracy in %, 500 puzzles per set; iterations to halt over halted puzzles). Pat.: patience rule; Arm.: Armijo backtracking on E.

Task	Model (rule)	Acc. ID	Acc. OOD	Halted ID	Halted OOD	It. ID (med/p90)	It. OOD (med/p90)
4×4	SFPRM (Pat.)	97.8	65.6	98.2	73.4	13 / 18	19 / 45
4×4	Energy (Pat.)	80.2	21.4	47.0	28.0	49 / 67	50 / 67
4×4	Energy (Arm.)	82.4	17.2	93.6	81.8	61 / 100	68 / 114
6×6	SFPRM (Pat.)	99.8	89.2	99.8	90.6	11 / 13	19 / 37
6×6	Energy (Pat.)	0.0	0.0	2.4	1.0	54 / 97	67 / 118
6×6	Energy (Arm.)	0.0	0.0	62.0	30.4	88 / 117	102 / 122

Codebook variant. The codebook runs of Table 3 use the hidden-token energy model of Section E with the answer codebook on the visible tokens ($\kappa_{\mathrm{ans}} = 0.5$, $\beta_{\mathrm{ans}} = \sqrt{d}$ at the end of the schedule, prototypes initialized as orthonormal directions and kept at unit norm), the tied readout with a learned scale, the digit embedding tied to the prototypes, the β schedule $\rho = 8$ over the first $K = 8$ iterations applied to both memory terms (so the energy is only non-increasing after iteration 8, and the energy-increase counter of Section E is not meaningful during the schedule), the ranking loss with weight 0.1, margin 1 and $R = 2$ restarts per puzzle (batch 64×2) when enabled, and otherwise the training recipe of Section E; 2,000 steps on 4×4 and 4,000 on 6×6. On 6×6 only the variant with the ranking loss was trained, at learning rate 10^{-3}; a first run at 2×10^{-3} diverged after 500 steps (loss rising from 0.93 to 1.34) and was stopped. A further 4×4 ablation fixes β at its final value from the first iteration (with the ranking loss): it stays on the clue-only plateau for all 2,000 steps, with 65% ID and 49% OOD cell accuracy, no board solved and no puzzle halted within 128 iterations. The memory-MLP ablation of SFPRM uses 256 tied memories with a learned β in place of the GELU MLP in both nGPT layers. Evaluation adds a third halting rule, Armijo backtracking with the commitment requirement, and reports the commitment statistics at the Armijo-halted state. With the commitment requirement, board accuracy is 97.6% ID and 71.4% OOD for the codebook model (98.0% and 74.4% without it) and 93.6% and 61.6% with the ranking loss (94.6% and 62.8%). Figure 5 shows the training curves and halting distributions of the codebook variant against SFPRM, and its 6×6 spectrum and restart curves.

F. Numerical Verification of the Energy Statements

All checks use the model of Definition 8.1 with random weights in double precision: $T = 6$ tokens, $d = 16$, $H = 2$ heads of dimension 8, $K = 24$ memories, $\kappa_0 = 1$, signed head weights $\sigma = (0.3, -0.3)$, $\beta_k = 2$, $\kappa_M = 0.5$, $\beta_M = 3$, and the geodesic injection; inputs x are normalized embeddings of random clues. Scripts are included with the submission.

- **Proposition 8.2.** The closed-form drive (31)–(32) is compared with $-\mathrm{P}_w \nabla E$ from automatic differentiation on

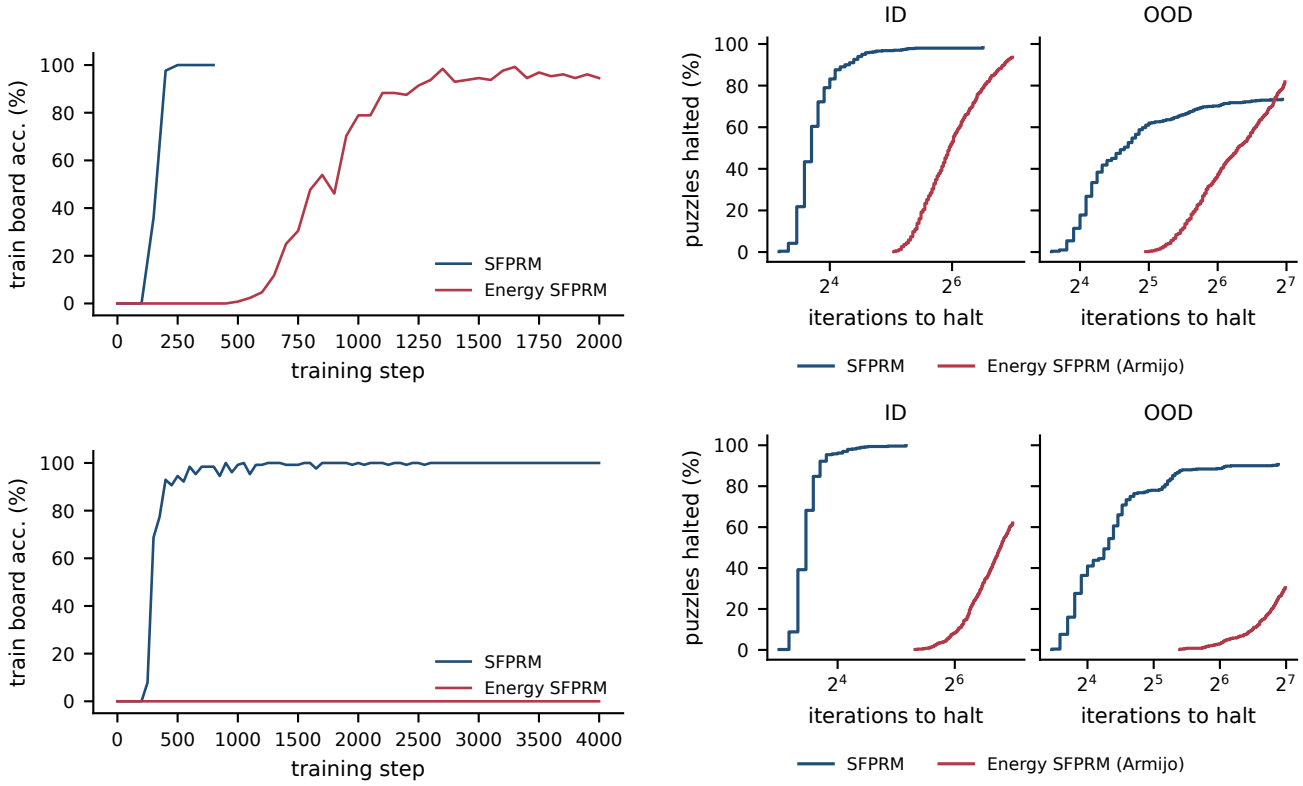


Figure 3. Training curves (left; train board accuracy, logged every 50 steps) and cumulative distribution of iterations to halt (right; SFPRM with the patience rule, the energy model with Armijo backtracking) on 4×4 (top) and 6×6 (bottom) Sudoku.

200 random states for the flat model and 100 for the hidden model. The coupling matrices J_{ts} of the flat model are assembled explicitly and compared with $J_{st}^{\top}$, and $\mathrm{P}_z \sum_s J_{ts} z_s$ with the attention part of the drive.

- **Proposition 8.4.** A stationary point is found by iterating (30) with $\eta = 0.2$ until $\|\mathrm{grad}\, E\| < 10^{-14}$ (765 iterations). The tangent Jacobian of F_E (automatic differentiation in an orthonormal basis of the tangent space) is compared with $\mathrm{I} - \mathrm{Hess}\, E$, where the Riemannian Hessian is $\mathrm{P}\, \nabla^2 E\, \mathrm{P} - \mathrm{M}_\mu$, and with central finite differences of step 10^{-6} along tangent directions. The Hessian eigenvalues lie in $[0.21, 1.90]$. For comparison, a random one-layer SFPRM map (scalar eigen learning rates 0.3, $\lambda = 0.5$) is iterated to a fixed point (residual $< 10^{-30}$) and its tangent Jacobian is analysed in the same way.

- **Theorem 8.3.** $\hat{L}$ is the largest Riemannian Hessian eigenvalue over 50 random states (2.23) and over the states and midpoints visited by 20 descent trajectories of 20 steps (3.05). 200 trajectories of 20 steps are then run at $\eta \in \{0.5, 1, 1.9, 3\}/\hat{L}$; at the three step sizes inside the window the energy never increased and the inequality of (i) held at every step (worst slack -5.9×10^{-8} at $1.9/\hat{L}$), while at $3/\hat{L}$ the energy increased in 54% of the steps.

- **Equation (34).** At a stationary point with Hessian condition number 6.5 and a linear loss $\ell(w) = \langle \omega, w \rangle$ with random ω, the adjoint system is solved exactly, by conjugate gradient, and by the truncated Neumann series $\eta \sum_{k<K} (\mathrm{I} - \eta\, \mathrm{Hess}\, E)^k$ with $\eta = 1/\lambda_{\max}$; the iteration counts to relative error 10^{-3} are 7 and 40 and to 10^{-6} they are 11 and 81. The resulting derivative of $\ell(w^\star)$ with respect to κ_0 is compared with central finite differences of step 10^{-5} through the re-solved stationary point.

- **Equation (35).** The nudged minimizer of $E + \beta\ell$ is found by the same iteration and (35) is evaluated one-sided and centered for $\beta \in \{10^{-1}, 10^{-2}, 10^{-3}\}$; the relative errors are 2.5, 0.34, 0.034 (one-sided, $O(\beta)$) and 0.050, 0.0010, 1.1×10^{-5} (centered, $O(\beta^2)$).

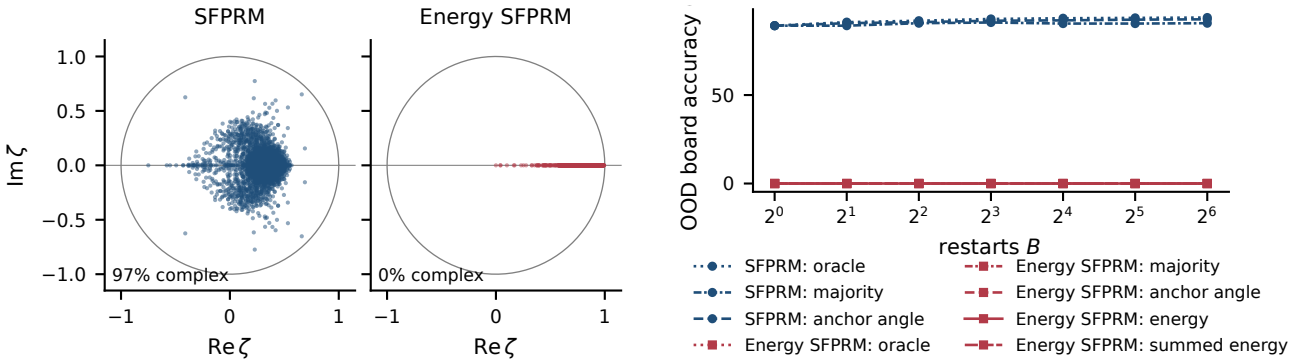


Figure 4. 6×6 pilot: Jacobian spectra at polished halted OOD states (left) and OOD board accuracy against the number of restarts (right), as in Figure 2, for the first energy model.

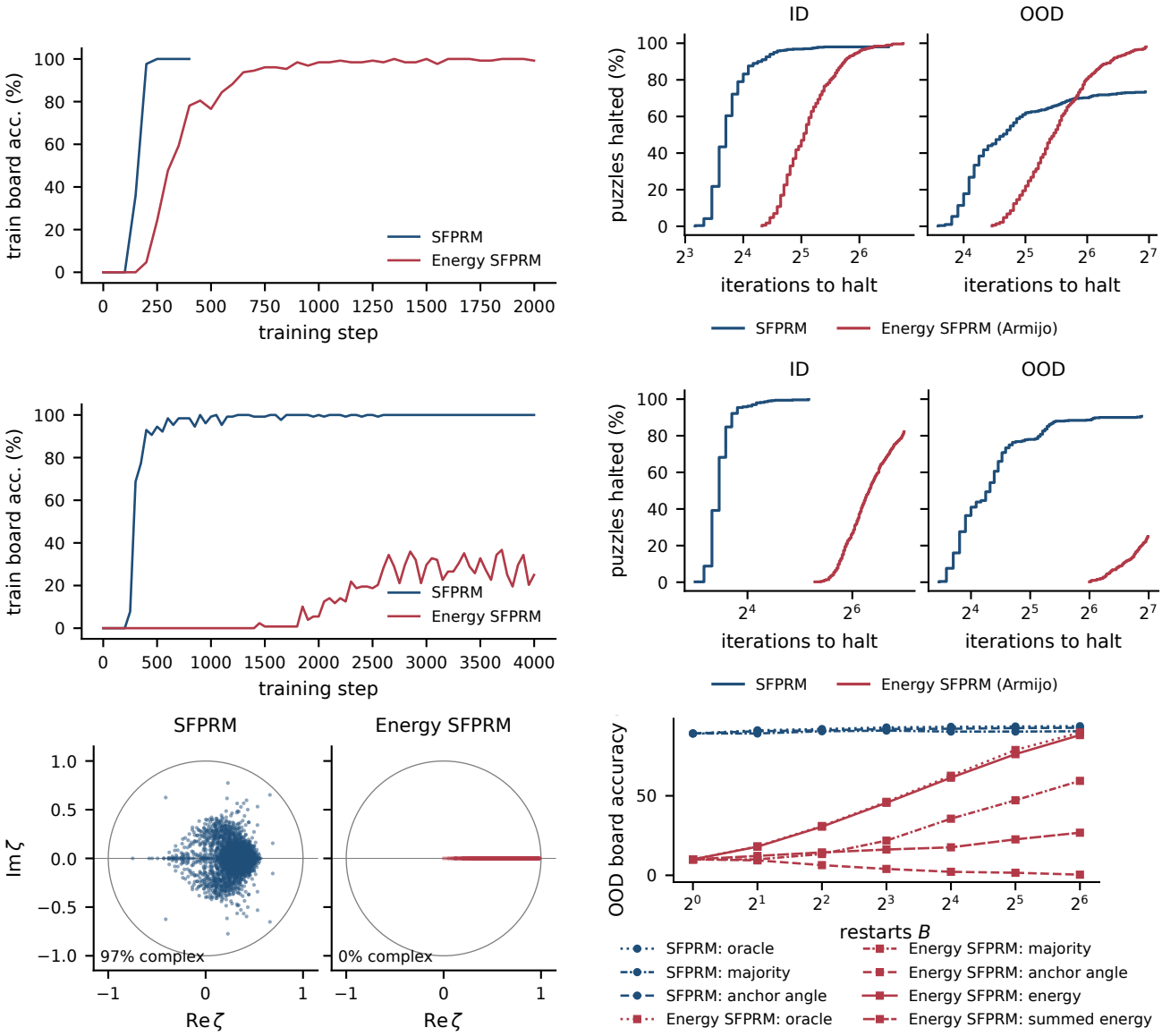


Figure 5. Codebook variant ("Energy SFPRM") against SFPRM. Top: 4×4 training curves and iterations to halt ("Codebook" column of Table 3). Middle and bottom: 6×6 training curves, iterations to halt, spectra at halted OOD states and OOD accuracy against restarts ("+rank" column).